

PAULEE GROUP · CENTER FOR QUANTUM TECHNOLOGY, KIST

Convolution Operation and Quantum Signal Processing

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Outline

1. Introduction to Quantum Information
2. Convolution Operation
3. Quantum Signal Processing

INTRO METHOD STRATEGY RESULTS ONGOING

01

Introduction to Quantum Information

Everything we do is three steps

Quantum information is an enormous field, but what we actually run in the lab generally has the same three steps.

STEP ONE

Prepare a state

The system is initialised into one known starting state.



STEP TWO

Apply a unitary

A quantum operation, which is to say an algorithm, evolves the state the way we want.



STEP THREE

Measure

One number comes back, and every conclusion has to be extracted from it.

$$|\psi_0\rangle \xrightarrow{U} U|\psi_0\rangle,$$

$$\text{what we read out} = \langle\psi_0|U^\dagger O U|\psi_0\rangle.$$

Each box has a long story behind it, and each is a field of its own. On the theory side the main interest sits in the unitary U : it is the part we choose, and we design it on purpose so that the number produced by the third box answers the question we asked.

State preparation

A state of a d -level system is a ket, a column of complex amplitudes in a basis $\{|j\rangle\}$:

$$|\psi\rangle = \sum_{j=0}^{d-1} c_j |j\rangle = \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_{d-1} \end{pmatrix} \quad c_j \in \mathbb{C}.$$

The bra is its Hermitian conjugate, the row vector of conjugated amplitudes $\langle\psi| = (c_0^*, \dots, c_{d-1}^*)$, and the norm is the bra taken against the ket,

$$\langle\psi|\psi\rangle = \sum_j |c_j|^2 = 1,$$

which is the normalization condition.

Every quantity we can actually observe is a bra times a ket: an expectation value $\langle\psi|O|\psi\rangle$, or an overlap $|\langle\phi|\psi\rangle|^2$.

Multiply the ket by a phase, $|\psi\rangle \rightarrow e^{i\alpha}|\psi\rangle$, and the bra picks up the conjugate, $\langle\psi| \rightarrow e^{-i\alpha}\langle\psi|$, so every such product is unchanged:

$$\langle\psi|e^{-i\alpha} O e^{i\alpha}|\psi\rangle = \langle\psi|O|\psi\rangle, \quad |\langle\phi|e^{i\alpha}|\psi\rangle|^2 = |\langle\phi|\psi\rangle|^2.$$

Therefore we lose no generality by taking the first amplitude c_0 to be real and non-negative.

So a d -level state needs $2d - 2$ real numbers, $2d$ from the amplitudes minus the norm and the phase. A qubit is left with **two**, which is why the next slides draw it as a point on a sphere.

Expectation value

An observable is a Hermitian matrix $O = O^\dagger$, and the number the experiment reports is $\langle O \rangle_\psi = \langle \psi | O | \psi \rangle = \sum_{j,k} c_j^* O_{jk} c_k$, a row vector times a matrix times a column vector, which Hermiticity makes real.

For a single qubit the observables of interest are the Pauli matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and we write the row, the matrix and the column, then multiply.

With $|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ this gives

$$\langle + | X | + \rangle = \frac{1}{2} (1, 1) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1.$$

One shot gives an eigenvalue, many shots give the average

The trace is cyclic, so the same number can be written with the state moved to the outside:

$$\langle \psi | O | \psi \rangle = \text{Tr} (| \psi \rangle \langle \psi | O) = \text{Tr}(\rho O), \quad \rho = | \psi \rangle \langle \psi |.$$

The object ρ is the **density matrix**, which is what one half of an entangled pair becomes later in this chapter.

A single shot does not return the average.

Diagonalise the observable, $O = \sum_k \lambda_k |k\rangle \langle k|$. One shot returns **one eigenvalue** λ_k with the Born probability $p_k = | \langle k | \psi \rangle |^2$, and it leaves the system in the corresponding eigenvector $|k\rangle$, which is the collapse.

Averaging those outcomes gives

$\sum_k \lambda_k p_k = \langle \psi | O | \psi \rangle$: the expectation value **is** the shot-averaged outcome, reached with an error of order $1/\sqrt{N}$ after N shots. For Z the eigenvalues are ± 1 and $\langle Z \rangle = p_0 - p_1$.

Bloch sphere

A qubit state is a vector in a two-dimensional complex space,

$$|\psi\rangle = c_0|0\rangle + c_1|1\rangle \text{ with } |c_0|^2 + |c_1|^2 = 1.$$

Take expectation value of pauli matrix, $\langle\sigma_i\rangle = \langle\psi|\sigma_i|\psi\rangle$, so the three form the

Bloch vector:

$$\vec{r} = (\langle X\rangle, \langle Y\rangle, \langle Z\rangle).$$

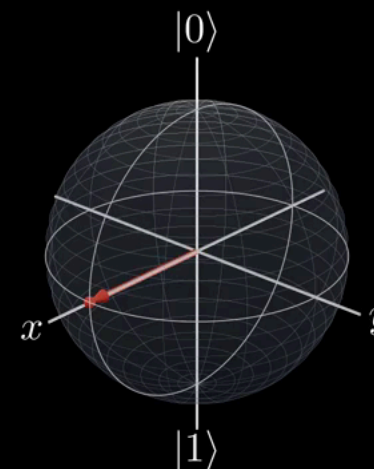
For example, $|+\rangle$,

$$\langle X\rangle = \frac{1}{2}(1 \quad 1) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1,$$

with $\langle Y\rangle = \langle Z\rangle = 0$, so $|+\rangle$ sits at $\vec{r} = (1, 0, 0)$.

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\vec{r} = (\langle X\rangle, \langle Y\rangle, \langle Z\rangle) = (1, 0, 0)$$



The two angles are the sphere's own coordinates

A ket has four real numbers. Normalisation removes one, and the global phase, which no expectation value can see, removes another. Only two real parameters are left, so by convention we write

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle.$$

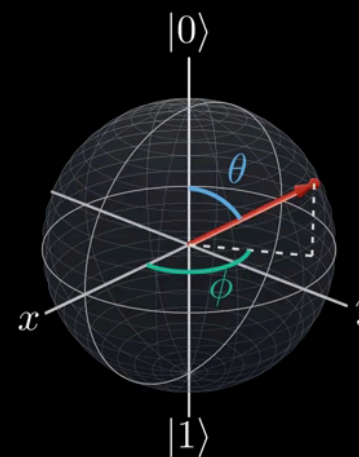
With $\langle X \rangle = 2 \operatorname{Re}(c_0^* c_1)$, $\langle Y \rangle = 2 \operatorname{Im}(c_0^* c_1)$ and $\langle Z \rangle = |c_0|^2 - |c_1|^2$ this gives

$$\vec{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta),$$

the spherical coordinates of a unit vector. θ is the polar angle from $|0\rangle$, ϕ the azimuth about z , and the half angle inside the ket is what makes the full angle θ appear on the sphere. Since $|\vec{r}| = 1$, **every ket sits on the surface**; the radius is the purity, $|\vec{r}|^2 = 2 \operatorname{Tr} \rho^2 - 1$.

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} \end{pmatrix}$$

$$\vec{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$$



$$\theta = 58^\circ$$

$$\phi = 120^\circ$$

The same products for the general ket, then the arrow with the polar angle θ (blue arc from the z axis) and the azimuth ϕ (green arc from the x axis) following it as both angles are swept.

Our case: prepare $|0\rangle$, read out along z



A laser pulse pumps the spin into $|0\rangle$, the unitary we designed acts, and a second laser pulse returns a photon count.

Strictly, the last step is a projective measurement on $|0\rangle$ rather than a measurement of Z , because a photon count is intrinsically positive. The observable is the projector $\Pi_0 = |0\rangle\langle 0|$, not an operator with eigenvalues ± 1 . The two are one affine step apart, since $\Pi_0 = \frac{1}{2}(I + Z)$:

$$P(0) = \langle \psi | \Pi_0 | \psi \rangle = \frac{1 + \langle Z \rangle}{2}, \quad \langle Z \rangle = 2P(0) - 1.$$

The mean count is linear in $P(0)$ between a bright and a dark reference level, so dividing by that contrast turns photons into $P(0)$. **After contrast normalisation the readout is effectively a Z measurement**, and $\langle Z \rangle$ is what every curve in this talk plots.

Schrodinger equation

The state obeys the Schrodinger equation (with $\hbar = 1$),

$$\frac{d}{dt}|\psi(t)\rangle = -i H |\psi(t)\rangle,$$

where H is the Hamiltonian, the matrix whose expectation value is the energy, and it is Hermitian, $H^\dagger = H$, because energies are real. If H does not depend on time this is a linear equation with constant coefficients, so an exponential solves it,

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle, \quad U(t) = e^{-iHt} = \sum_{n=0}^{\infty} \frac{(-iHt)^n}{n!}.$$

It is easy to see that $U^\dagger U = e^{+iH^\dagger t} e^{-iHt} = e^{+iHt} e^{-iHt} = I$, and the same in the other order. So the Schrodinger equation generates a unitary operation, and the norm of the state is preserved.

Hamiltonian and unitary operation

To implement the gate operation we need, we have to engineer the Hamiltonian. The Hamiltonian is always drift plus control, $H(t) = H_0 + H_c(t)$: what the system does on its own, plus the pulse we add.

For example, the NV centre has its own Hamiltonian, and we can apply a microwave pulse to it, which is effectively an oscillating magnetic field:

$$\begin{aligned} H_0 &= D S_z^2 + \gamma B_{\parallel} S_z, \\ H_c(t) &= \Omega(t) \cos(\omega t + \varphi) S_x. \end{aligned}$$

Here D is the zero-field splitting, $\gamma B_{\parallel}$ the Zeeman shift, $\Omega(t)$ the drive amplitude, ω its carrier and φ its phase, which in the rotating frame selects the rotation axis; the sequences later use $\varphi = \pi/2$, a y drive.

Now $H(t_1)$ and $H(t_2)$ no longer commute, and the time-dependent Schrodinger equation has no closed form in general; solving it efficiently is a challenge of its own and out of scope for today. Briefly, one can step it forward like the Euler method,

$$|\psi(t + dt)\rangle = |\psi(t)\rangle - i H(t) |\psi(t)\rangle dt,$$

and the product of many such small steps is the unitary U .

Hamiltonian on Bloch Sphere

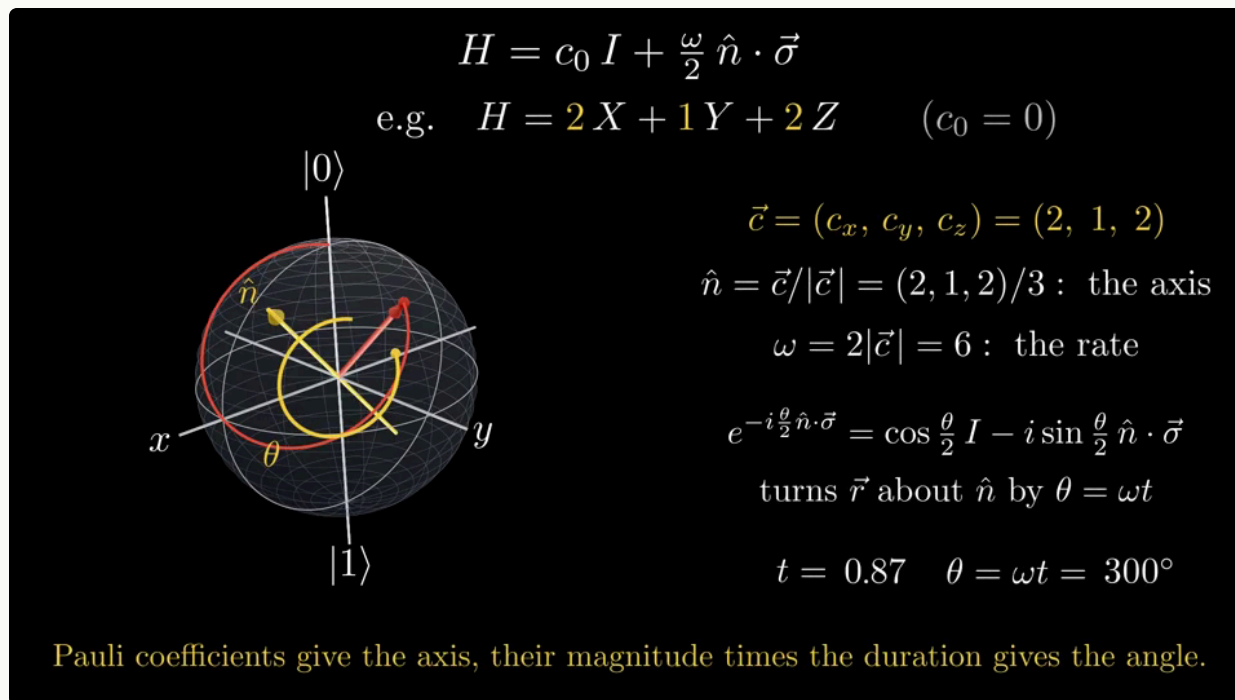
A 2×2 Hermitian matrix has four real parameters, so it is always a combination of Pauli matrices,

$$H = c_0 I + \frac{\omega}{2} \hat{n} \cdot \vec{\sigma}, \quad \hat{n} = \frac{\vec{c}}{|\vec{c}|}, \quad \omega = 2|\vec{c}|,$$

with $\vec{c} = (c_x, c_y, c_z)$ the Pauli coefficients and $c_0 I$ a global phase. Then

$$e^{-i\frac{\theta}{2}\hat{n}\cdot\vec{\sigma}} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \hat{n} \cdot \vec{\sigma}, \quad \theta = \omega t,$$

rotates the Bloch vector about $\hat{n}$ by θ : the Pauli coefficients give the axis, their magnitude times the duration gives the angle. A resonant pulse $H = \frac{\Omega}{2} Y$ turns the state about y by Ωt ; free precession $H = \frac{\delta}{2} Z$ about z by δt .



$H = 2X + Y + 2Z$ has $\vec{c} = (2, 1, 2)$, so $\hat{n} = (2, 1, 2)/3$ and $\theta = 6t$; the state $|0\rangle$ precesses on a cone about $\hat{n}$. Then $\frac{\Omega}{2} Y$ turns $|0\rangle$ toward $|+\rangle$ by Ωt , and $\frac{\delta}{2} Z$ turns $|+\rangle$ about z by δt .

So, prepare $|0\rangle$, apply a pulse U , and read $P(0)$.

Take one spin-1 transition as the qubit. In the rotating frame, after the rotating-wave approximation and with the drive phase along y , the Hamiltonian is constant:

$$H = \frac{\delta}{2} Z + \frac{\Omega}{2} Y.$$

The detuning δ is where the field enters and $\Omega = \gamma B_1$ is the drive strength.

The exponent is written in the Pauli basis, so the time evolution is a rotation and both of its parts are read off the coefficients:

$$U = e^{-iHt} = \exp\left(-\frac{i}{2}\Omega_R t \hat{n} \cdot \vec{\sigma}\right), \quad \hat{n} = \frac{(0, \Omega, \delta)}{\Omega_R}.$$

The **axis** is the coefficient vector, the **angle** is its length $\Omega_R = \sqrt{\Omega^2 + \delta^2}$ times the time. Drive off means precession about z .

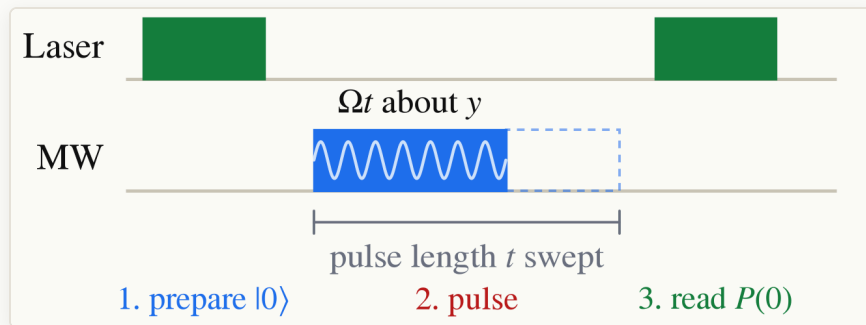
Rabi: calibrating the rotation angle

Set the pulse on resonance, $\delta = 0$, so

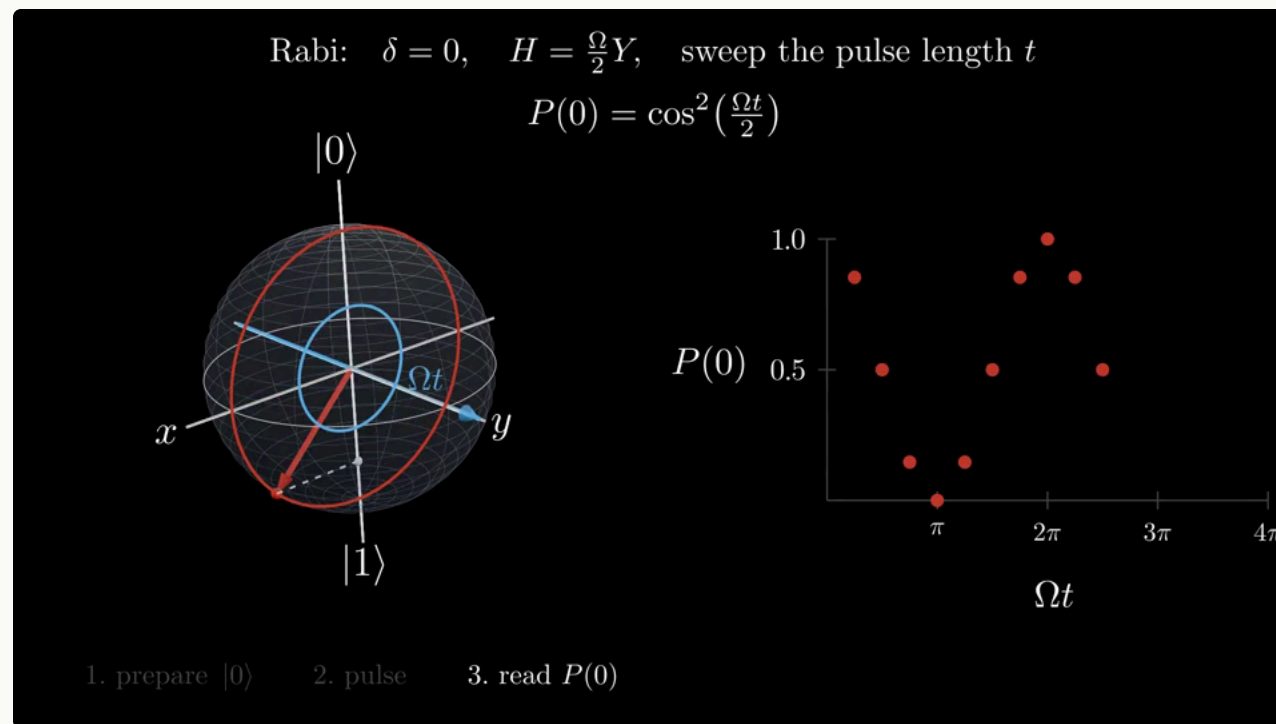
$H = \frac{\Omega}{2}Y$ and the pulse is a rotation about y by the angle Ωt :

$$P(0) = |\langle 0 | e^{-i\Omega Y t/2} | 0 \rangle|^2 \\ = \cos^2\left(\frac{\Omega t}{2}\right).$$

Sweep the pulse length and the fluorescence oscillates at Ω .



One microwave pulse, its length t swept.



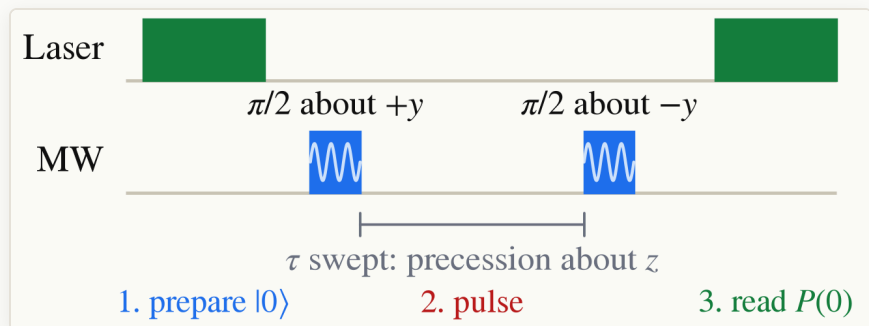
The state turns about y through the arc Ωt , the dashed line projects it onto z , and each read-out adds one point to $P(0)$.

Ramsey

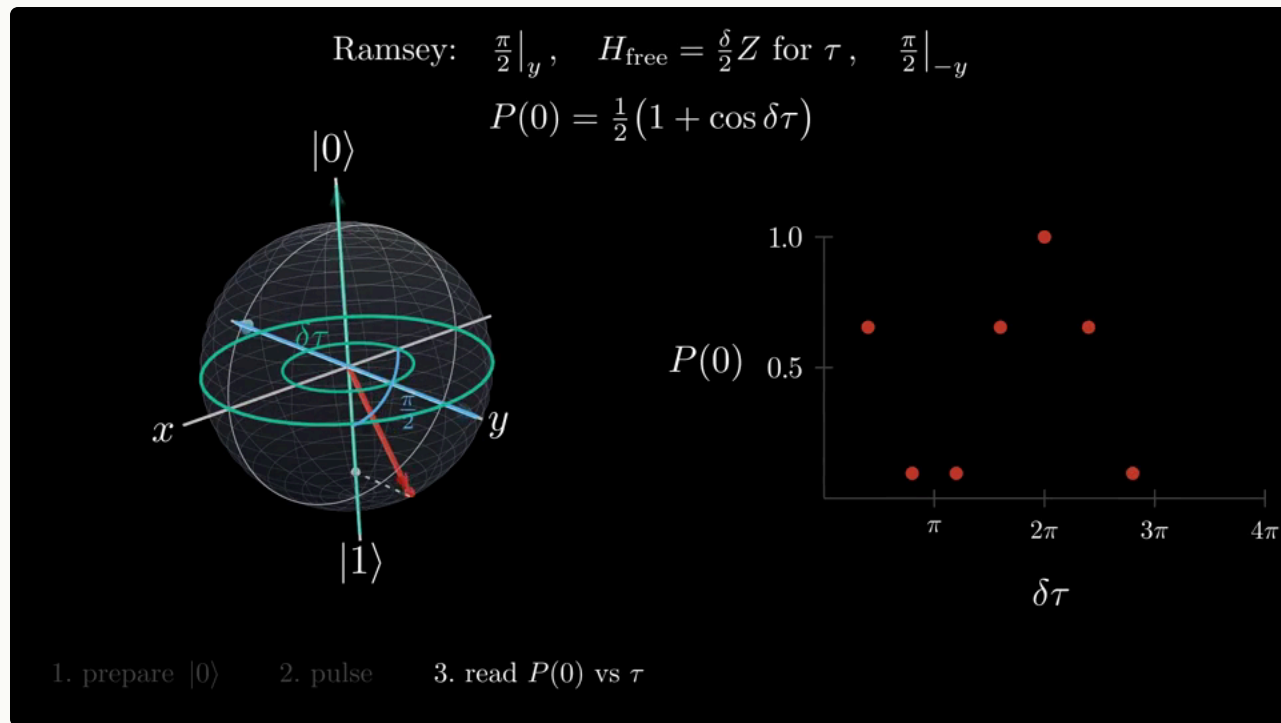
A $\pi/2$ pulse about the y axis gives the $|+\rangle$ state. Then, with the drive off, $\Omega = 0$, only $H = \frac{\delta}{2}Z$ acts, so the state precesses about z by $\delta\tau$. Finally apply a second $\pi/2$ pulse, about $-y$; the resulting population depends on that phase:

$$P(0) = \frac{1}{2} \left[1 + \cos(\delta\tau) \right].$$

The frequency is the detuning itself. A field shift moves δ , and the fringe counts it.



Two fixed $\pi/2$ pulses, the wait τ swept.

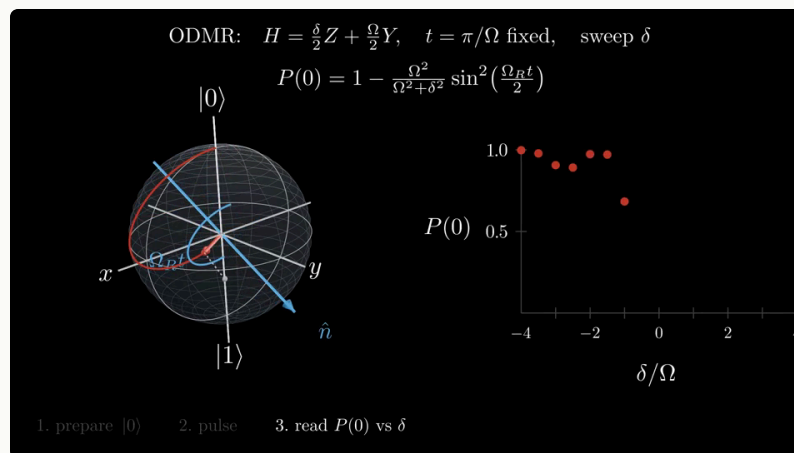


For each τ : $\pi/2$ about y , precession about z by $\delta\tau$, $\pi/2$ about $-y$, then the projection onto z gives one point.

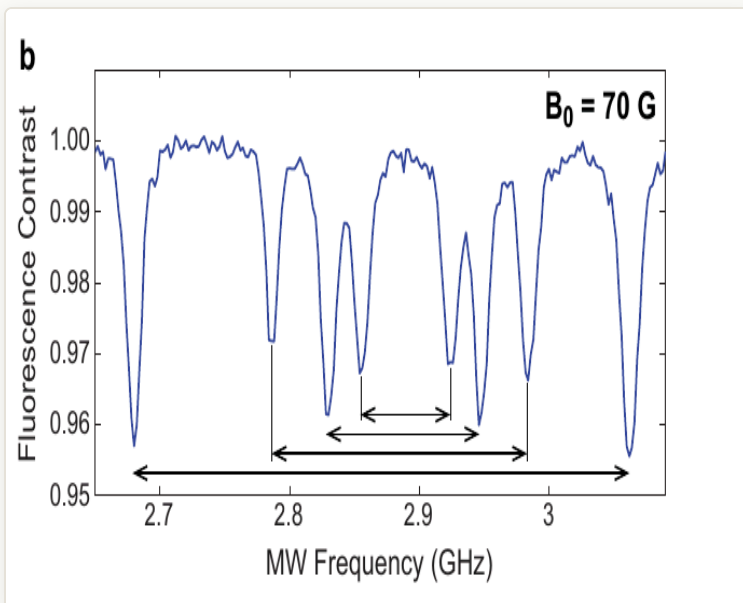
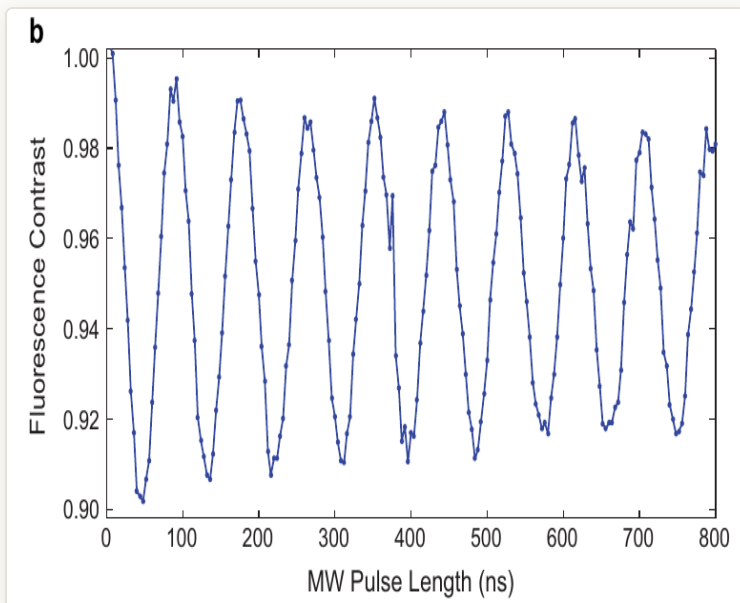
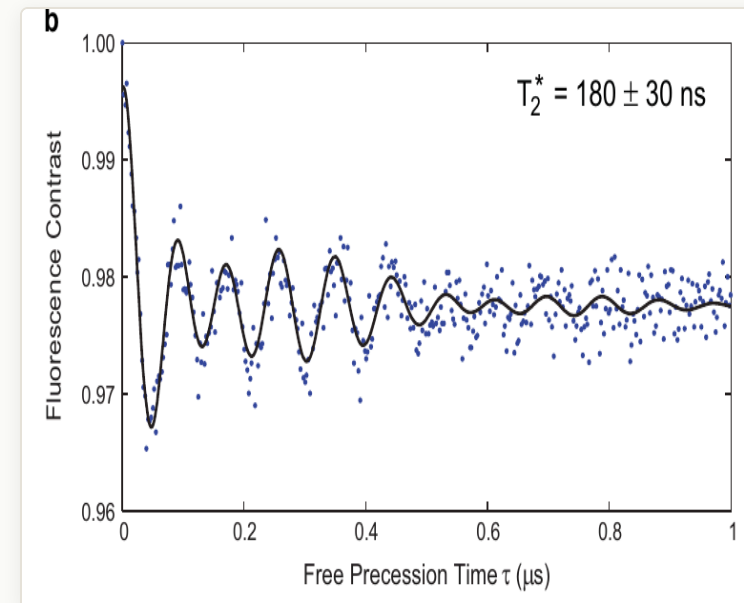
Fix the pulse at the calibrated length $t = \pi/\Omega$ and sweep the carrier instead. So on resonance, $\delta = 0$, we have $\Omega t = \pi$, a π pulse. Off resonance, the rotation axis $\hat{n} = (0, \Omega, \delta)/\Omega_R$ tilts toward z and the rotation is faster:

$$P(0) = 1 - \frac{\Omega^2}{\Omega^2 + \delta^2} \sin^2\left(\frac{\Omega_R t}{2}\right).$$

The resulting signal is approximately a sinc function.



The axis tilts as δ is stepped across $[-4\Omega, 4\Omega]$, and one π pulse per step draws out the dip and its side lobes.

*ODMR**Rabi**Ramsey*

Measured on an NV ensemble (Pham thesis, Ch. 1, Figs. 1.4, 1.5, 1.7). Two things remain to discuss. One is how the response is lost as time goes on, and the other is the source of the response.

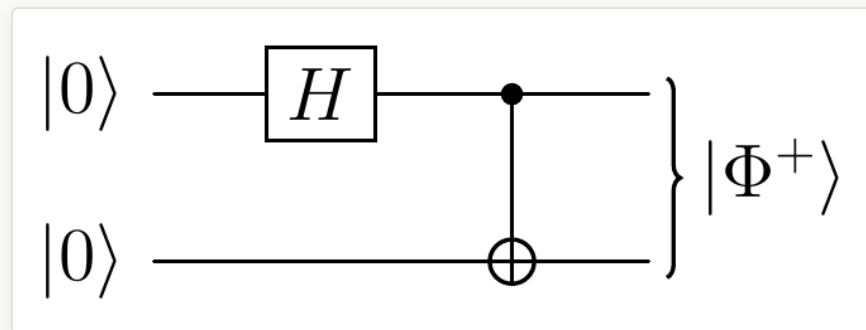
Entanglement

We start from a product state, $|\psi_0\rangle = |0\rangle|0\rangle$. A Hadamard acts on the first qubit and leaves the second one alone,

$$(H \otimes I) |00\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)|0\rangle.$$

The CNOT is **conditional**: it flips the second qubit only when the first one is $|1\rangle$. It gives the maximally entangled Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle).$$



Any product state can be written as $(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$, which would need $ad = bc = 0$ together with $ac = bd \neq 0$. No product of two kets reproduces the Bell state.

Mixed state

Suppose our interest is the first qubit. Every measurement on it is $\text{Tr}(\rho_1 O)$, with the other qubit traced out,

$$\rho_1 = \text{Tr}_2 |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2} I.$$

This has rank two, while a ket gives

$\rho = |\psi\rangle\langle\psi|$ of rank one: no ket describes the first qubit alone, so we call it **mixed**. Every

Pauli matrix is traceless, so

$\langle\sigma_i\rangle = \frac{1}{2} \text{Tr} \sigma_i = 0$ and the arrow sits at the centre, $|\vec{r}_1| = 0$. The length is the purity, $\text{Tr} \rho^2 = \frac{1}{2} (1 + |\vec{r}|^2)$: 1 on the surface, 1/2 at the centre.

Entanglement with the outside looks like mixing inside our qubit: the more entangled, the shorter the Bloch vector.

Two qubits: an interaction builds the Bell state

$$|\psi_0\rangle = |0\rangle |0\rangle$$

$$(H \otimes I) |\psi_0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) |0\rangle$$

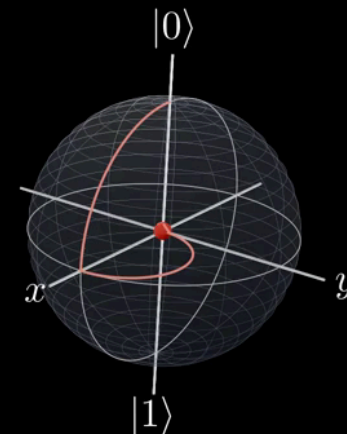
$$\text{CNOT} : |\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$\rho_1 = \text{Tr}_2 |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2} I$$

$$\langle\sigma_i\rangle = \text{Tr}(\frac{1}{2} I \sigma_i) = \frac{1}{2} \text{Tr} \sigma_i = 0 \Rightarrow |\vec{r}_1| = 0$$

$$\langle X \rangle = 0.00 \quad \langle Y \rangle = 0.00 \quad \langle Z \rangle = 0.00$$

$$|\vec{r}_1| = \sqrt{\langle X \rangle^2 + \langle Y \rangle^2 + \langle Z \rangle^2} = 0.00$$



The reduced Bloch vector of qubit 1: from $|0\rangle$ to $|+\rangle$ under the Hadamard, then to the origin during the CNOT, with $\langle X \rangle$, $\langle Y \rangle$, $\langle Z \rangle$ read off the partial trace.

Interaction is the source of entanglement.

The CNOT came from an interaction, and in the NV centre the hyperfine coupling to a nearby nucleus has the same conditional form, with $\hat{H}_0 = \omega_L \hat{I}_z$ and $\hat{H}_1 = \hat{H}_0 + A\hat{I}_z + B\hat{I}_x$:

$$\begin{aligned}\hat{H} &= |0\rangle\langle 0| \otimes \hat{H}_0 + |1\rangle\langle 1| \otimes \hat{H}_1, \\ |+\rangle|n\rangle &\rightarrow \frac{1}{\sqrt{2}} (|0\rangle U_0 |n\rangle + |1\rangle U_1 |n\rangle), \\ |\vec{r}_e| &= |\langle n| U_1^\dagger U_0 |n\rangle|.\end{aligned}$$

The nucleus precesses about a different axis in each branch, since $H_0 \neq H_1$, so the electron's Bloch radius is the overlap of the two branches. There are many independent nuclei, so multiply their overlaps, giving $M(t) \simeq e^{-(t/T_2^*)^2}$.

Three names for the same shrinking, NV at room temperature.

- **T_2^* , free precession.** The Ramsey decay set by the static spread of local fields, ≈ 180 ns for one NV and ≈ 500 ns for an ensemble.
- **T_2 , spin echo.** A π pulse refocuses whatever was static, so only the motion of the bath survives, giving 397 ± 5 μ s.
- **T_1 , relaxation.** Phonons flip the spin and the populations decay, typically ~ 6 ms.

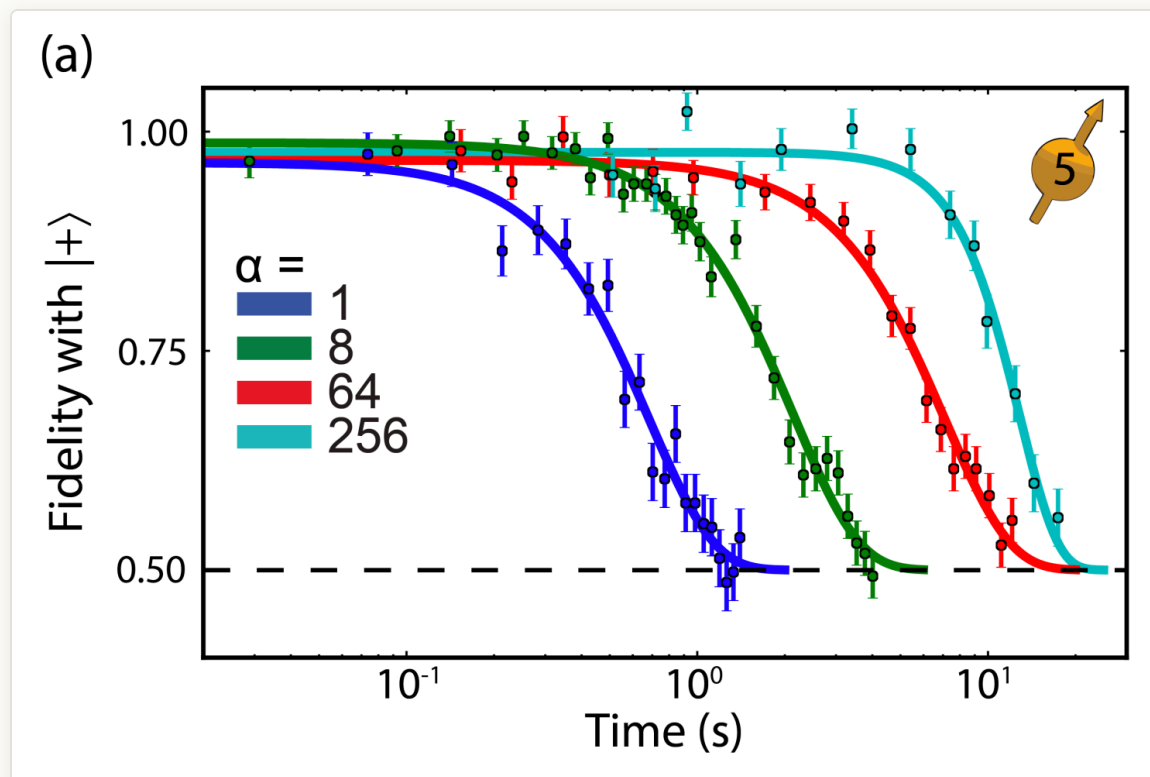
Natural-abundance ^{13}C caps T_2^* near 1 μ s; 99.7% ^{12}C material exceeds 10 μ s (Pham thesis, Ch. 1; conditional form from Taminiau et al., 2012).

The decay has a name: T_2^*

The shrinking radius of the last slide is exactly what a Ramsey experiment measures. The free-precession signal fades within a time T_2^* , set by the static spread of local fields: each nucleus shifts the electron's precession frequency by its own hyperfine coupling, and the many shifted precessions dephase when we add them up,

$$M(t) \simeq e^{-(t/T_2^*)^2}.$$

At room temperature $T_2^* \approx 180$ ns for a single NV and ≈ 500 ns for an ensemble, capped by the 1.1% natural abundance of ^{13}C (Pham thesis, Ch. 1).



A measured version of the same decay: the fidelity of one spin with $|+\rangle$, which falls to the 0.5 floor as the Bloch radius shrinks to zero. The bare curve $\alpha = 1$ decays first, like the T_2^ above; the longer ones are the same spin under the dynamical decoupling of the next slides. The time axis is logarithmic, and its scale is that spin's own.*

Dynamical decoupling: make the conditional operation unconditional

$$H = |0\rangle\langle 0| \otimes H_0 + |1\rangle\langle 1| \otimes H_1 \quad \Longrightarrow \quad U(t) = e^{-iHt} = |0\rangle\langle 0| \otimes U_0(t) + |1\rangle\langle 1| \otimes U_1(t).$$

For one ^{13}C near an NV centre,

$$H_0 = \omega_L I_z \text{ and}$$

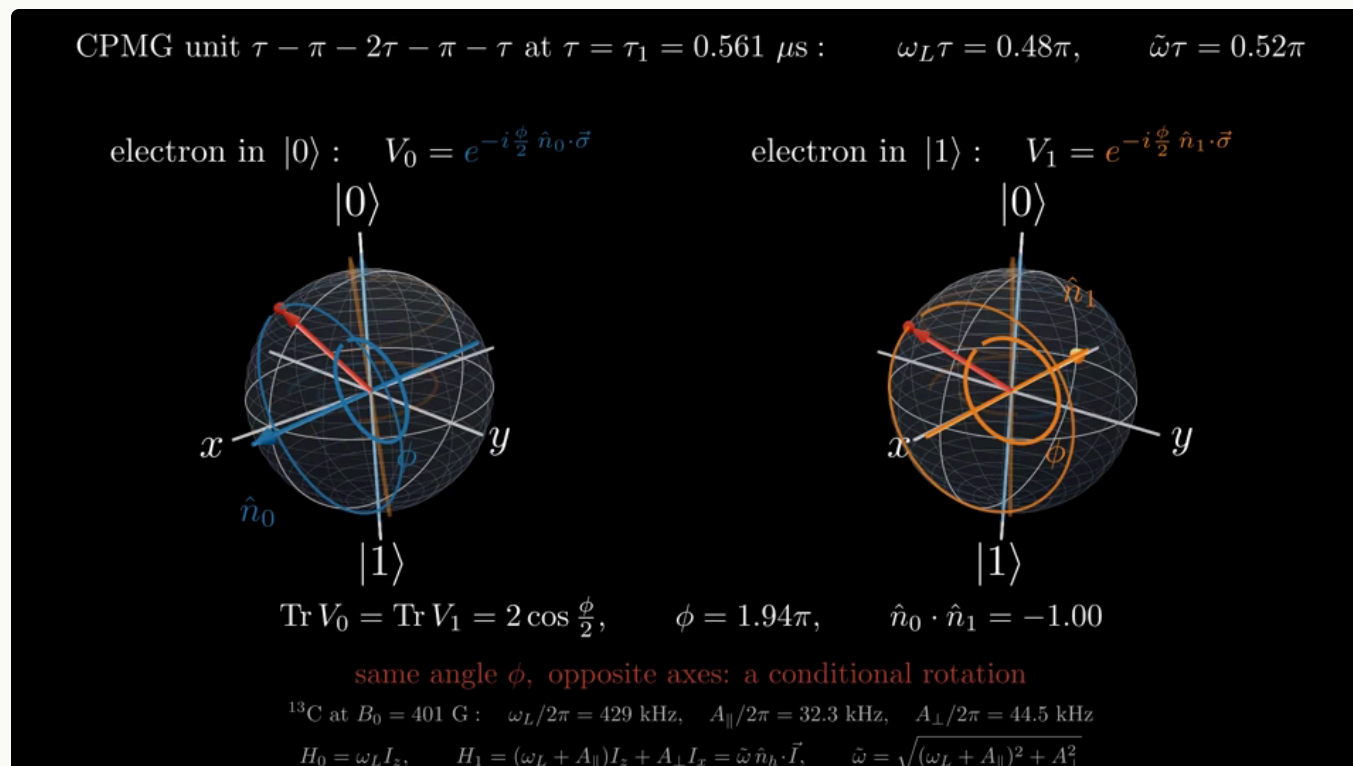
$$H_1 = (\omega_L + A_{\parallel}) I_z + A_{\perp} I_x. \text{ A } \pi \text{ pulse swaps the two blocks, so the}$$

CPMG unit $\tau - \pi - 2\tau - \pi - \tau$

interleaves them:

$$V_0 = U_0 U_1 U_1 U_0, \quad V_1 = U_1 U_0 U_0 U_1.$$

Both are unitary, so the only question is whether the two rotations agree. At a generic τ they do: $V_0 = V_1$, the electron sees no difference between its branches, and $M = 1$. Only special spacings break that, as the next slide shows.



One unit at a generic spacing, $\tau = 0.280 \mu\text{s}$, with the electron in $|0\rangle$ (left) and in $|1\rangle$ (right) driving the nucleus along the same path. The four factors collapse to one rotation per branch, by the same angle $\phi = 0.999\pi$ about axes with $\hat{n}_0 \cdot \hat{n}_1 = 0.999$.

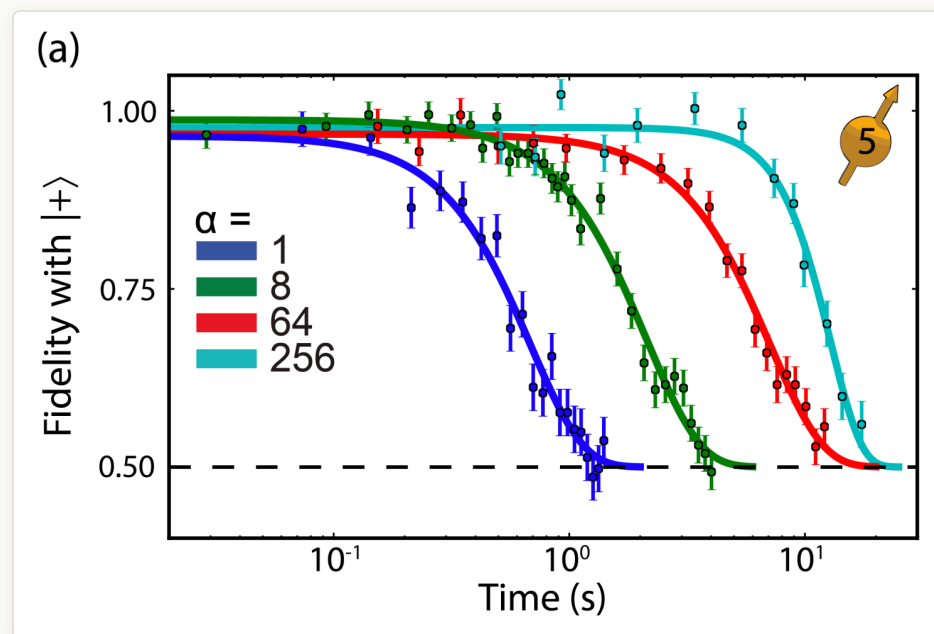
The read-out is one number: how well the two branches agree

Prepare the electron in $|+\rangle$, leave the nucleus unpolarised, $\rho_n = I/2$, and run N units. Tracing the nucleus out leaves the electron coherence $\langle 0|\rho_e|1\rangle = \frac{1}{2}M$, so a read-out along x returns

$$P_x = \frac{1}{2} + \text{Re} \langle 0|\rho_e|1\rangle = \frac{1+M}{2}, \quad M = \frac{1}{2} \text{Tr} (V_0^N V_1^{\dagger N}), \quad |M| \leq 1.$$

That $\frac{1}{2}$ is the trace normalisation of the unpolarised nucleus, so M is an average of $V_1^{\dagger N} V_0^N$. If the operation is **unconditional**, $V_0 = V_1$, then $M = 1$ exactly and no signal is lost.

The electron Bloch vector has length $|M|$ and purity $\text{Tr} \rho_e^2 = \frac{1}{2}(1 + |M|^2)$, so staying unconditional keeps the state pure. More pulses therefore buy a longer coherence time $T_2^{(N)}$.



More decoupling, longer coherence: the decay to the 0.5 floor moves from about 1.5 s at $\alpha = 1$ out to about 25 s at $\alpha = 256$, as the unconditional picture predicts.

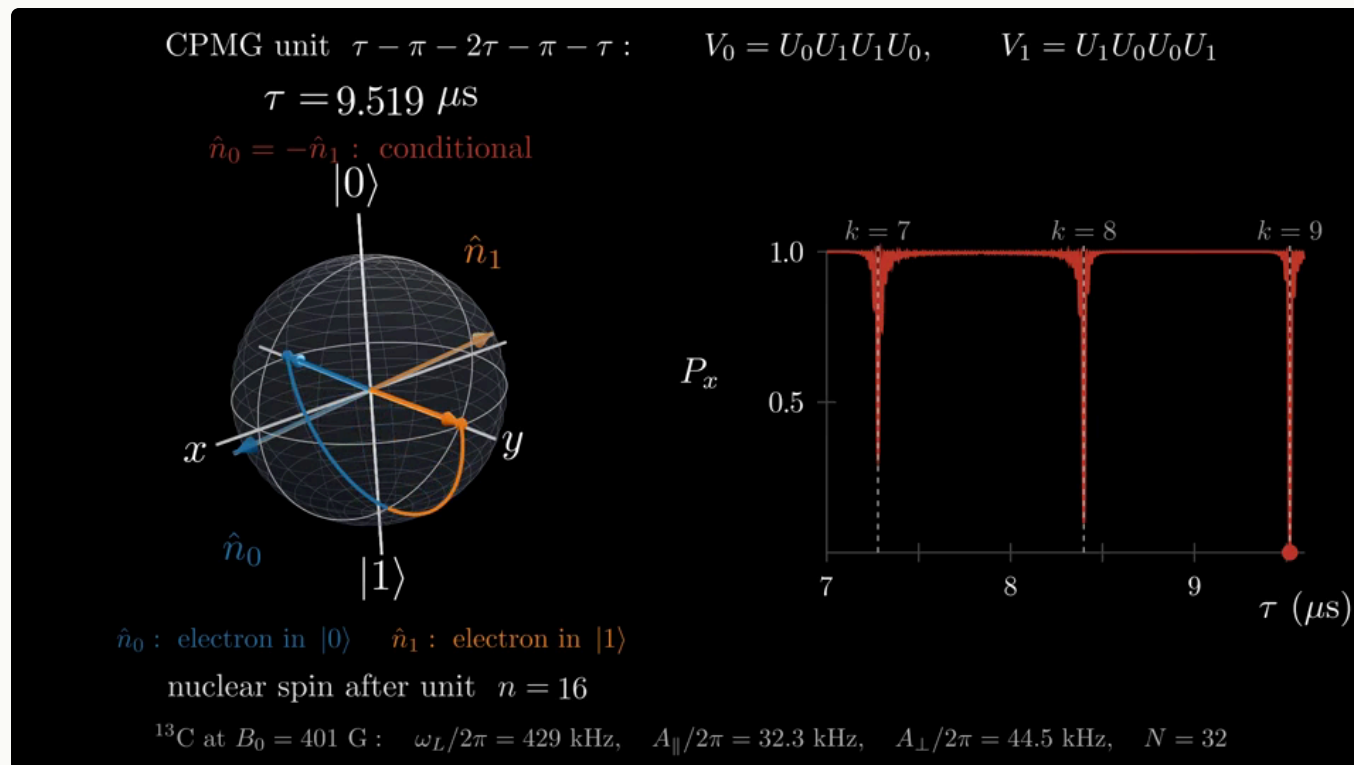
Same angle, different axis: the dips that name one nucleus

With $A = U_0(\tau)$ and $B = U_1(\tau)$, cyclicity of the trace gives $\text{Tr}(ABBA) = \text{Tr}(A^2B^2)$, that is $\text{Tr } V_0 = \text{Tr } V_1$, so both branches turn by the **same angle** ϕ and only the axes $\hat{n}_0, \hat{n}_1$ can differ:

$$M = 1 - (1 - \hat{n}_0 \cdot \hat{n}_1) \sin^2 \frac{N\phi}{2}.$$

Each nucleus has its own $A_{\parallel}$, hence its own comb of spacings where the axes anti-align. That comb is a fingerprint: it names the nucleus.

$$\tau_k = \frac{(2k-1)\pi}{2\omega_L + A_{\parallel}}.$$



Spin 3 of the paper's Table I ($B_0 = 401 \text{ G}$, $N = 32$ units): the axes coincide, open up as τ nears τ_7, τ_8, τ_9 , and anti-align there, where M plunges. The nucleus starts in $|0\rangle$, and at the resonance the two branches run round opposite halves of one great circle, antipodal after 16 units. The curve $P_x(\tau)$ is computed for $\rho_n = I/2$, so it does not depend on that choice.

What to carry out of chapter 1

- **An experiment is three steps.** We prepare a known state, apply a unitary, and measure.
- **The unitary is the part we design on purpose.**
- **The response is not the sequence alone.** It also carries the configuration of the system.

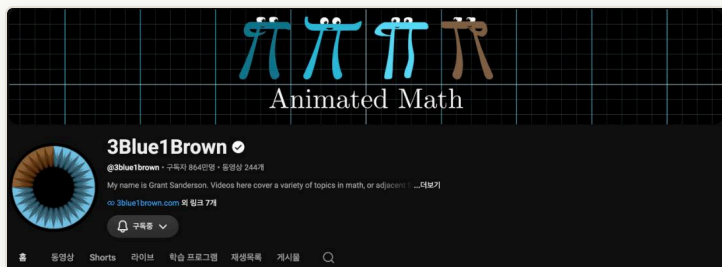
The measured curve is the designed response combined with the structure of the system, and that combination is a convolution.

02

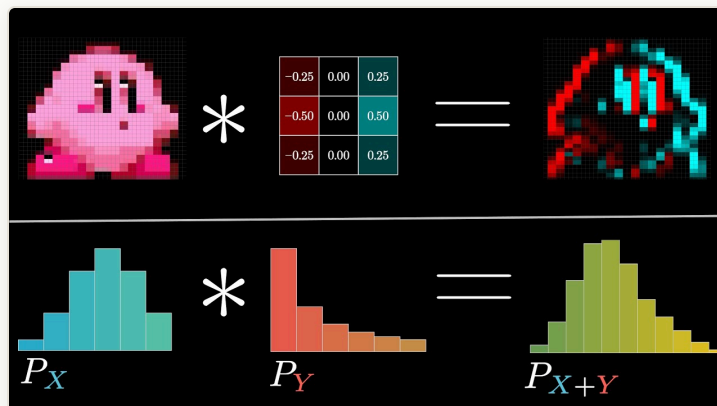
Convolution Operation

Credits: this chapter follows 3Blue1Brown

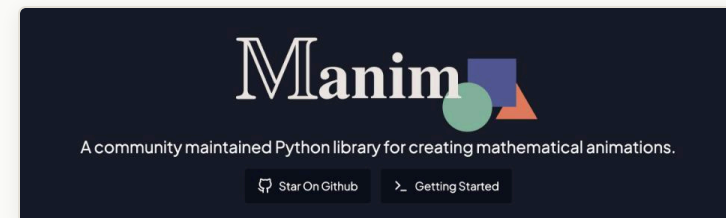
Everything in this chapter is told the way Grant Sanderson tells it in *But what is a convolution?* on the 3Blue1Brown channel, and the dice and polynomial clips are cut from his published source code (MIT licence). The Bloch-sphere and QSP clips of the other chapters were made with the community edition of Manim, the animation engine written for that channel.



The 3Blue1Brown channel,
youtube.com/@3blue1brown.



"But what is a convolution?" (2022),
youtu.be/KuXjwB4LzSA.

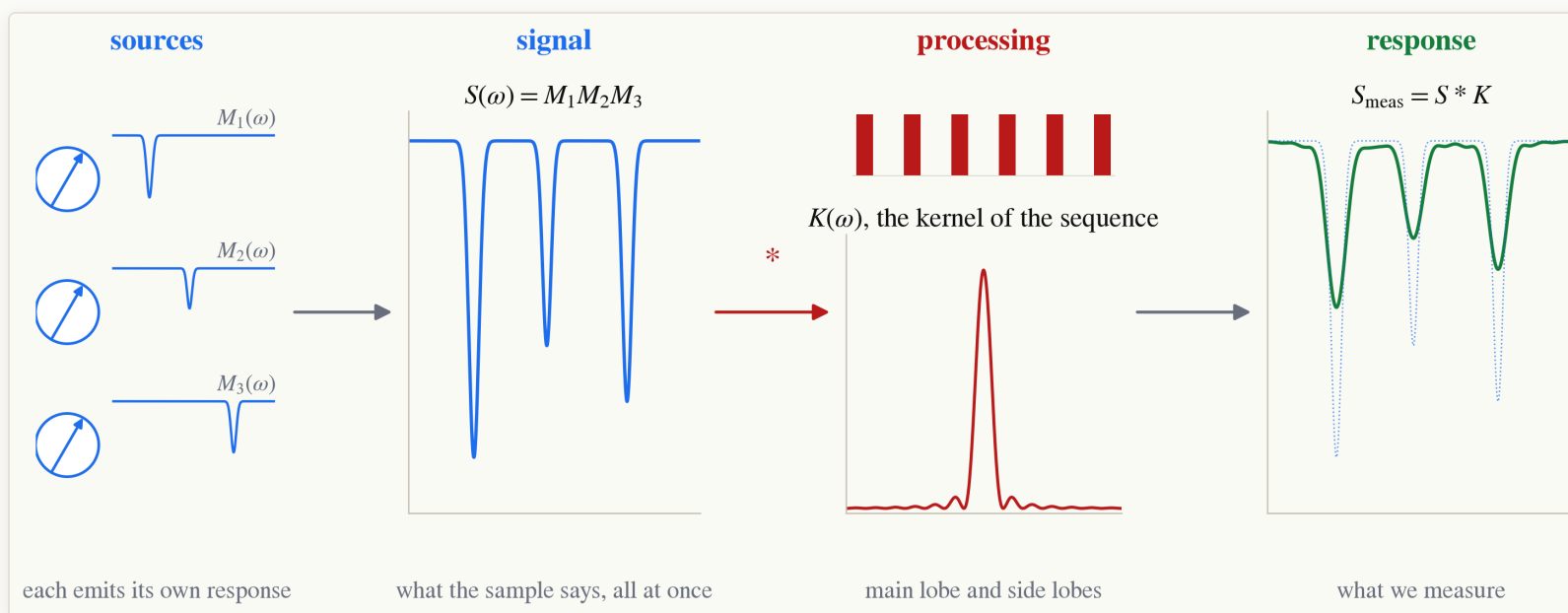


Manim Community Edition, manim.community.

The response is already convolved

The **sources**, the spins in the sample, respond to our input, the **signal**, through the **processing** we chose. I call the output the **response**, spectrum included, and it is convolved: it carries the configuration of the sample as well as the sequence we ran,

$$S_{\text{meas}}(\omega) = \int d\omega' S(\omega') K(\omega - \omega') \equiv [S * K](\omega).$$



*Three sources emit their own dips $M_j(\omega)$; together they are the signal $S(\omega)$; the sequence acts as a kernel $K(\omega)$; the screen shows $S * K$. The definition looks complicated, so let me first show why convolution is natural, starting from two dice.*

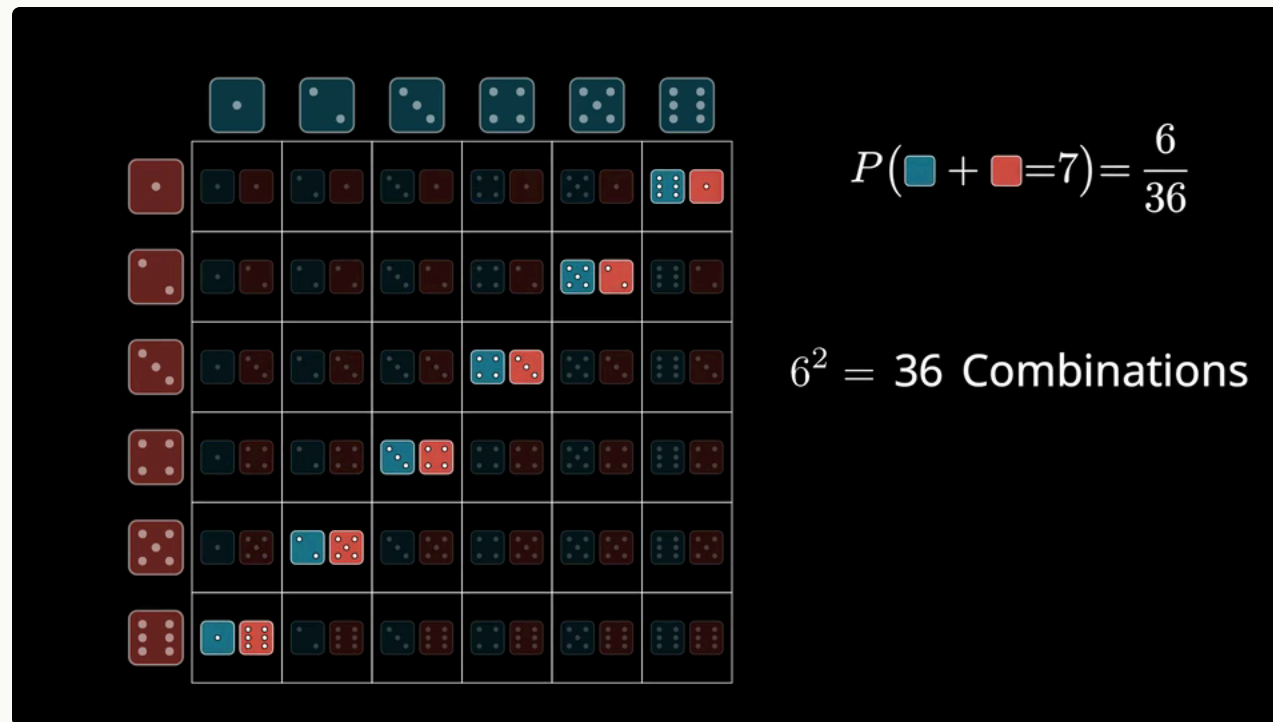
Two dice: the table of outcomes

Roll two independent dice X and Y and ask for the probability of their sum, $Z = X + Y$. The natural picture is a table: 36 cells, one per pair (X, Y) , each with weight $\frac{1}{36}$.

The event $Z = n$ happens whenever $X = k$ **and** $Y = n - k$, for any k , so the cells with a fixed sum lie on one anti-diagonal of the table:

$$P(Z = n) = \sum_k P(X = k) P(Y = n - k).$$

Counting the anti-diagonal for $n = 7$ gives six cells, so $P(Z = 7) = 6/36$, the largest of all.



The 6×6 table of pairs, and the anti-diagonals $X + Y = n$ marched one by one. Each diagonal is one entry of the distribution of the sum.

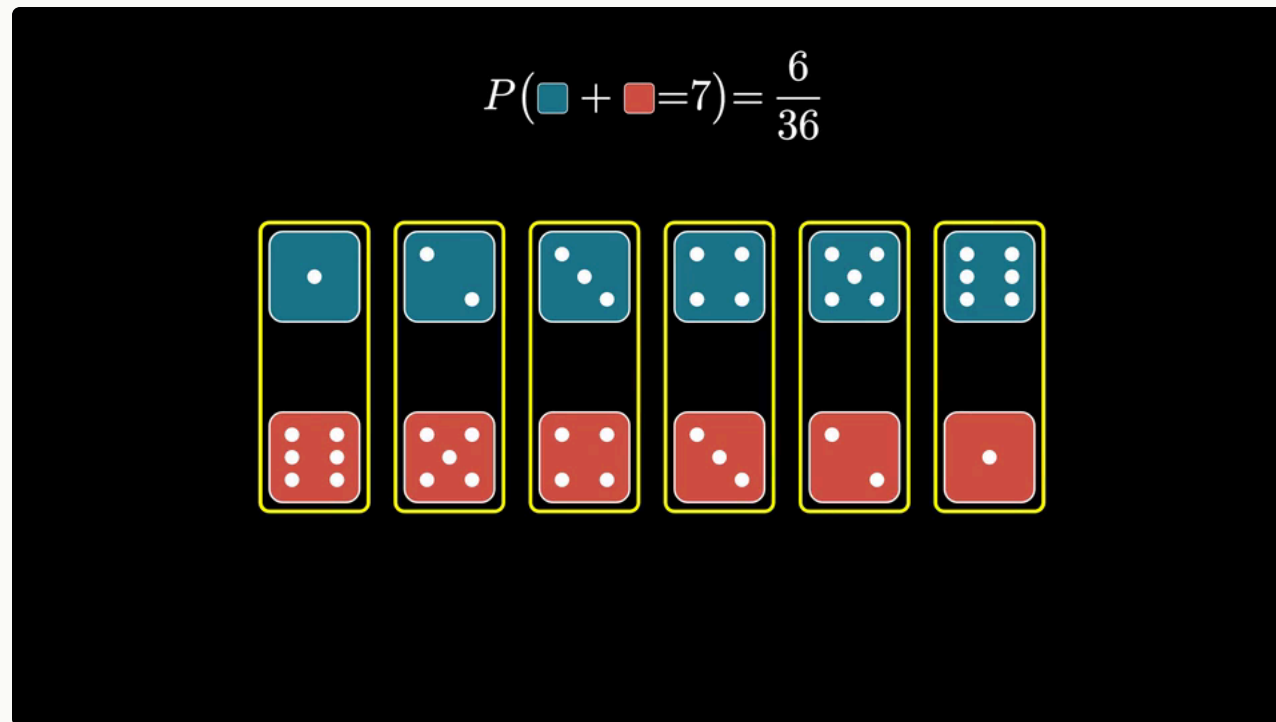
The same sum, seen as flip and slide

Write the two distributions as lists. Reverse one of them, slide it along the other, and at each displacement multiply the entries that face each other and add the products. That single number is $P(Z = n)$ for that displacement.

$$(f * g)[n] = \sum_k f[k] g[n - k].$$

The minus sign is the **flip** and n is the **slide**. The only rule is that the two indices add up to the output index, since $k + (n - k) = n$; everything else is bookkeeping.

Two lists of length 6 overlap at 11 displacements, the sums 2 to 12.



The second row is reversed and slid across the first. The pairs that line up are exactly the anti-diagonal of the table, now read as a moving window.

Uneven dice: the same picture with weights

Now load the dice, so the faces are no longer equally likely. Nothing changes in the bookkeeping: the same anti-diagonal is summed, only each cell now carries the product of two unequal weights,

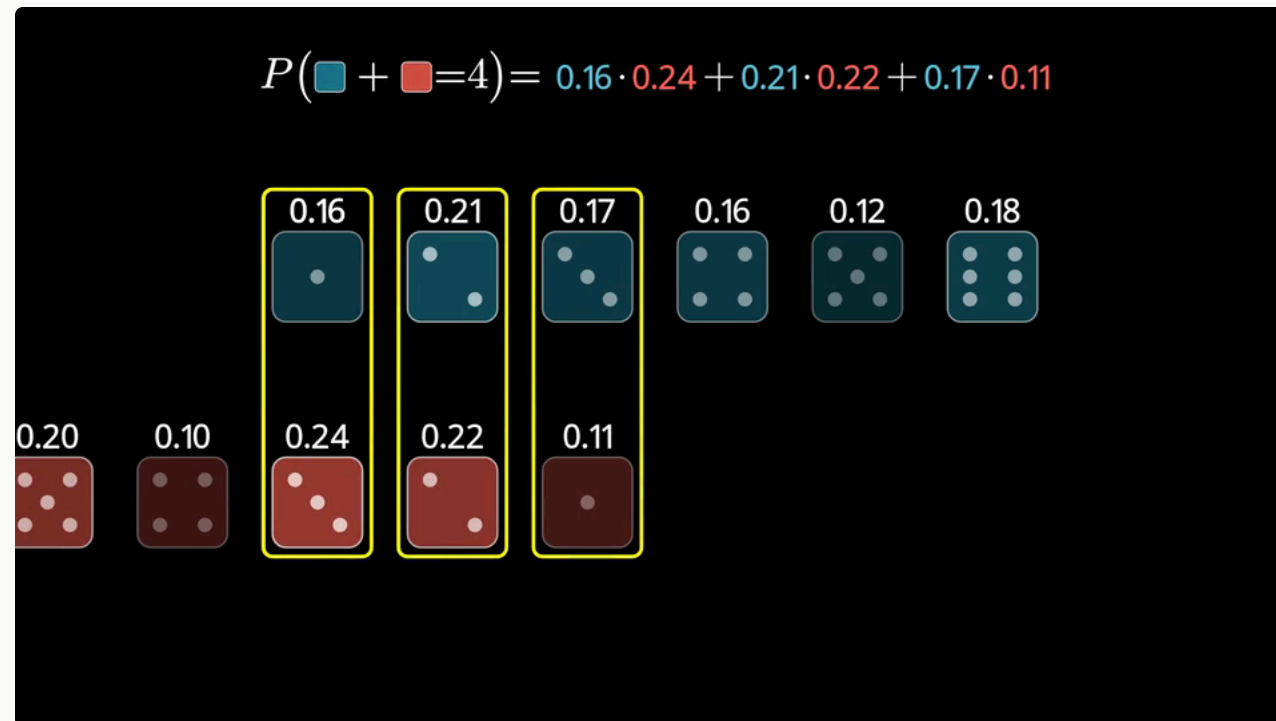
$$P(Z = n) = \sum_k P(X = k) P(Y = n - k) = [P_X * P_Y](n).$$

For example, the clip reads off

$$P(Z = 4) = 0.16 \cdot 0.24 + 0.21 \cdot 0.22 + 0.17 \cdot 0.11.$$

Nobody chose to convolve anything.

Adding two independent random variables **is** a convolution of their distributions.



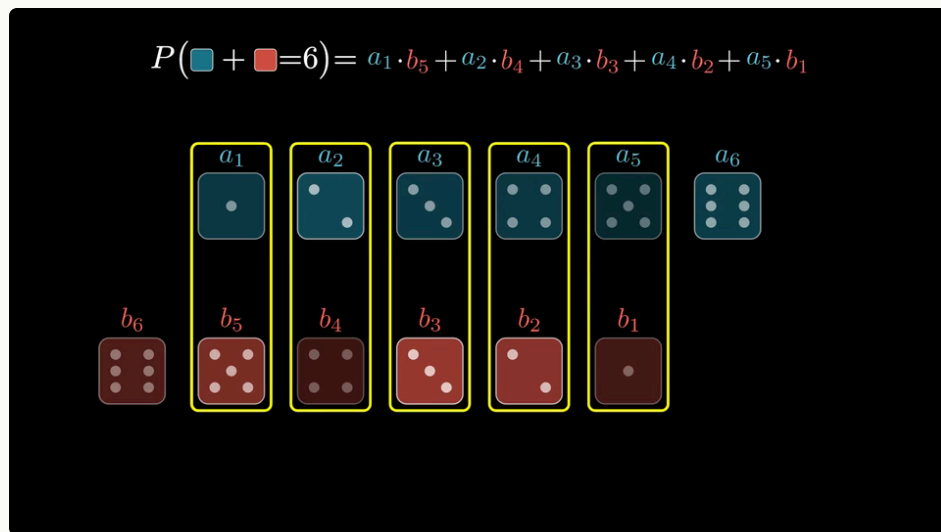
Loaded dice: the same table and the same anti-diagonals, now weighted cell by cell.

Convolution of (a_i) and (b_i)

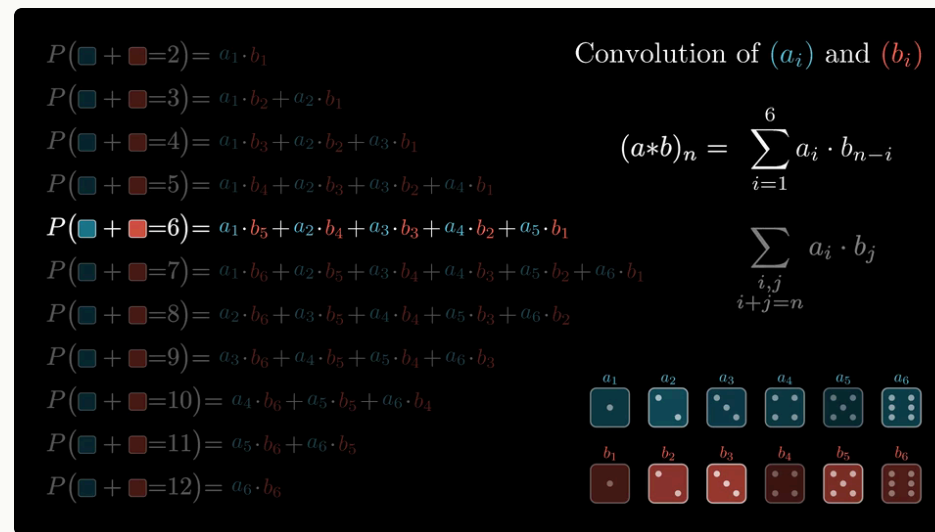
Replace the weights by symbols, $P_X(i) = a_i$ and $P_Y(j) = b_j$. The sliding window collects the pairs whose indices add up to n , and that is the whole definition:

$$(a * b)_n = \sum_{i+j=n} a_i b_j = \sum_i a_i b_{n-i}.$$

Up to now we have only added two independent random variables. The convolution was not a choice; it is what addition looks like in the distributions.



Symbolic weights, one window position per output entry.



The same products as a table, then the formula.

Forget the dice: any two lists convolve the same way

Take two plain lists, $(1, 2, 3)$ and $(4, 5, 6)$, with no probabilities attached. Reverse one, slide it, multiply what faces each other, add:

$$(1, 2, 3) * (4, 5, 6) = (4, 13, 28, 27, 18).$$

At displacement $n = 3$ the window holds $2 \cdot 6 + 3 \cdot 5 = 27$. Two lists of length 3 overlap at 5 displacements, so the output has length 5.

This is the discrete definition, and the continuous one is the same sentence with an integral in place of the sum:

$$(f * g)(x) = \int f(y) g(x - y) dy.$$

$$(1, 2, 3) * (4, 5, 6) = (4, 13, 28, \quad)$$

$$1 \cdot 6 + 2 \cdot 5 + 3 \cdot 4$$

a_0	a_1	a_2
1	2	3
6	5	4
b_2	b_1	b_0

One list reversed and slid across the other; each frame is one output entry.

A product of polynomials is a convolution of coefficients

Use the same numbers as coefficients and multiply two polynomials, which is an ordinary **pointwise** product of two functions of x :

$$\begin{aligned} (1 + 2x + 3x^2)(4 + 5x + 6x^2) \\ = 4 + 13x + 28x^2 + 27x^3 + 18x^4. \end{aligned}$$

The coefficient of x^n collects every pair of powers that adds up to n , and that is the flip-and-slide sum again:

$$c_n = \sum_{i+j=n} a_i b_j = (a * b)[n].$$

Multiplying by x^j **shifts** the coefficient list by j , so a product of polynomials is a weighted sum of shifted copies, which is what a convolution is.

$$(1, 2, 3) * (4, 5, 6) = (4, 13, 28, 27, 18)$$

$$(1 + 2x + 3x^2)(4 + 5x + 6x^2) = 4 + 13x + 28x^2 + 27x^3 + 18x^4$$

	1	$2x$	$3x^2$
4	4	$8x$	$12x^2$
$5x$	$5x$	$10x^2$	$15x^3$
$6x^2$	$6x^2$	$12x^3$	$18x^4$

Each power of x collects one diagonal of the table of pairwise products, and the diagonals are exactly the sums $i + j = n$.

The same picture for two curves: flip, slide, integrate

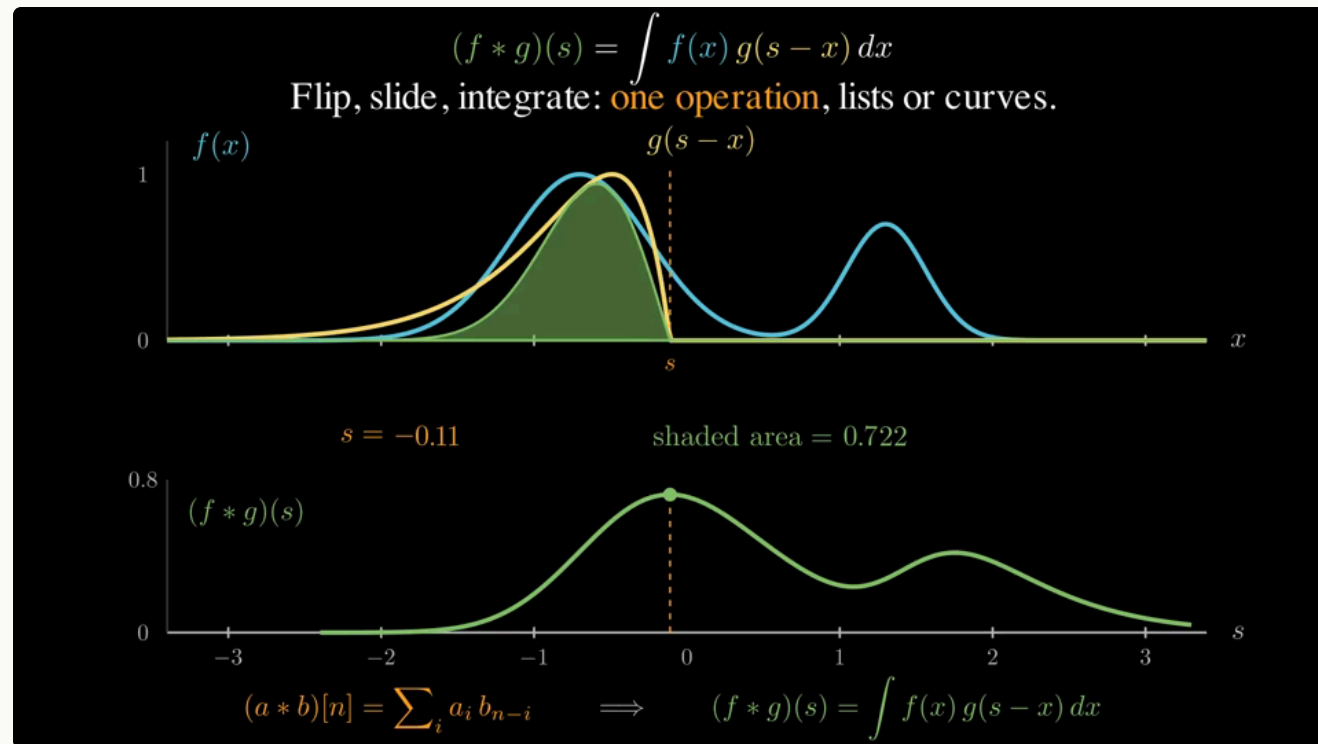
For two curves the definition reads the same, with the sum replaced by an integral:

$$(f * g)(x) = \int f(y) g(x - y) dy.$$

Hold f still, reverse g , slide it by x , multiply point by point, and add up over a continuum.

The polynomial product just seen is the discrete case of this picture, since a list is a function on the integers.

If f and g are two probability densities, then $f * g$ is the density of their sum.



The clip calls the displacement s : $f(x)$ in blue stands still, the flipped kernel $g(s - x)$ in yellow slides across it, their product is shaded, and the shaded area is traced below as $(f * g)(s)$.

One identity: convolve here, multiply there

$$\widehat{f * g} = \hat{f} \cdot \hat{g} \quad \Longleftrightarrow \quad \widehat{f \cdot g} = \hat{f} * \hat{g}.$$

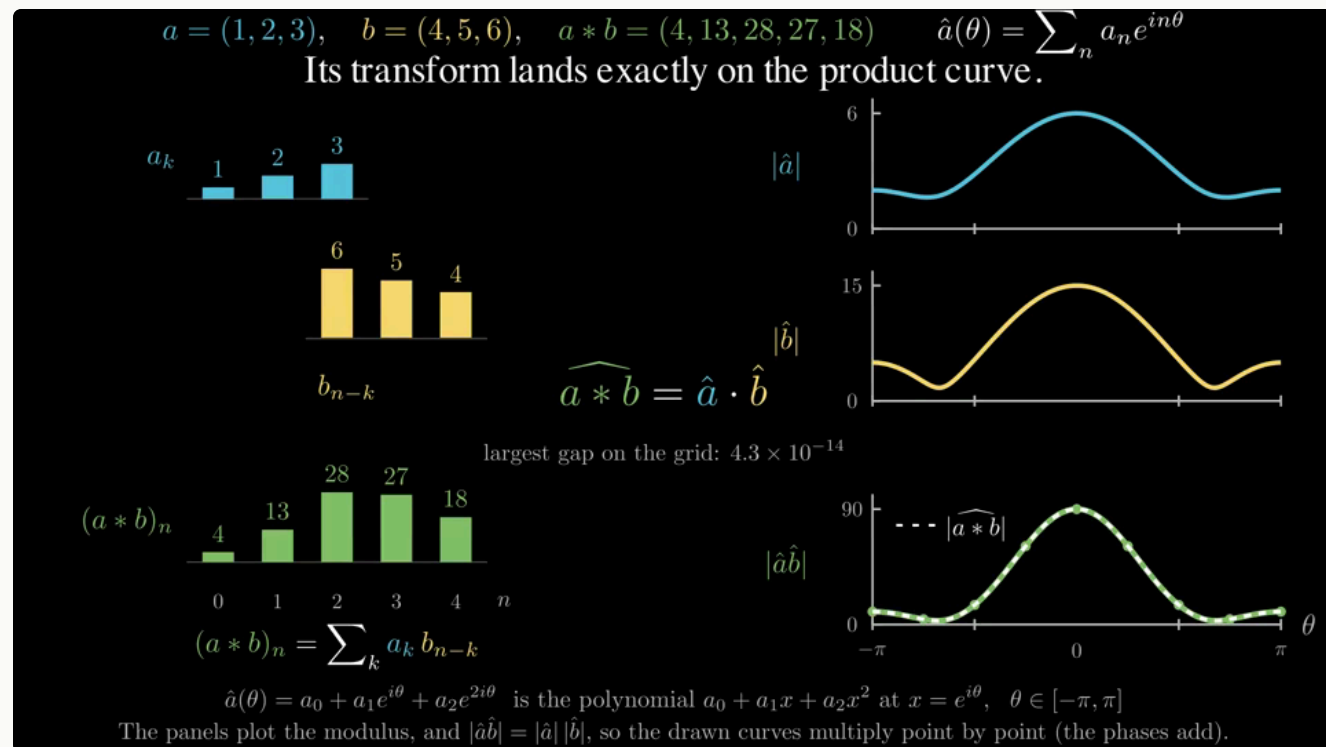
A convolution treats all positions alike, so it commutes with shifts, and the complex exponentials $e^{i\omega x}$ are its eigenfunctions. In that basis the operation is **diagonal**, and diagonal means multiply entry by entry.

Read the other way, a product of two records in time is a convolution of their spectra.

A trigonometric polynomial

$F(\theta) = \sum_n c_n e^{in\theta}$ is the polynomial slide with $x = e^{i\theta}$, so multiplying two

such responses **convolves their coefficient lists**. That is the case we live in.



Convoluting $a = (1, 2, 3)$ with $b = (4, 5, 6)$ on the left, multiplying $|\hat{a}(\theta)|$ by $|\hat{b}(\theta)|$ on the right: the dashed curve, the transform of the convolution, lands exactly on the product. The panels plot moduli, since $|\hat{a}\hat{b}| = |\hat{a}| |\hat{b}|$.

Our case: many nuclei, and the responses multiply

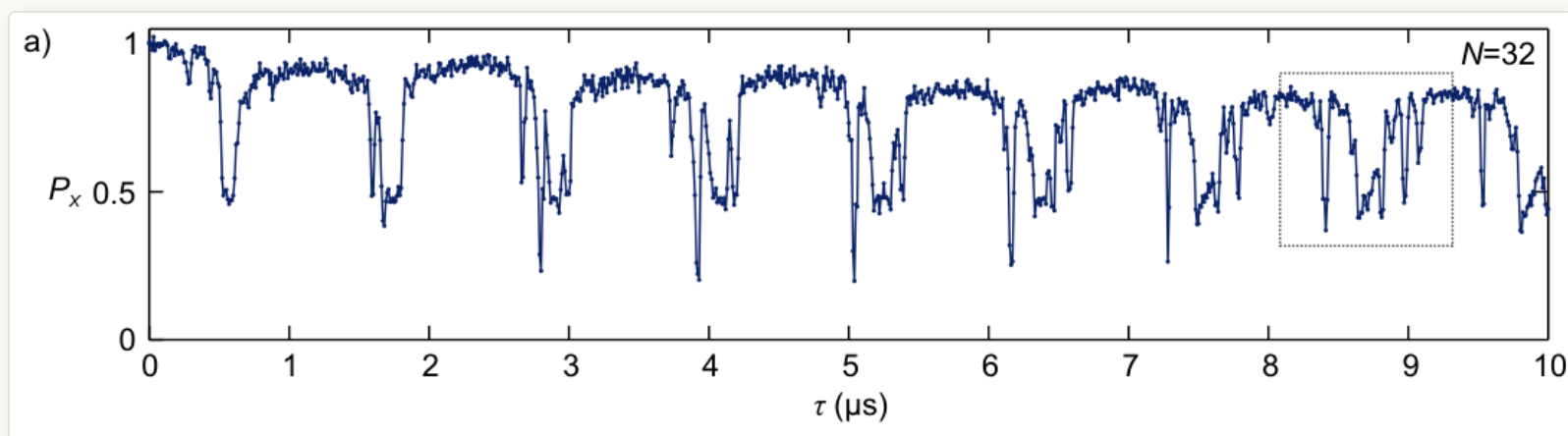
The electron is prepared in $|+\rangle$, a dynamical decoupling sequence runs, and it is read out at azimuth φ , where V_0 and V_1 are the propagators the **nuclear** spin experiences with the electron in $|0\rangle$ and in $|1\rangle$.

$$P(\varphi) = \frac{1}{2} + \frac{1}{2}|M| \cos(\varphi - \arg M), \quad M = \frac{1}{2} \text{Tr } V_0 V_1^\dagger$$

For several nuclei that do not interact with each other the trace factorizes.

$$M = \prod_j M_j, \quad M_j = \frac{1}{2} \text{Tr } V_0^{(j)} V_1^{(j)\dagger}$$

Each dip below is one nucleus telling the branches apart, the several combs multiply, and the plotted $P_x = \frac{1}{2}(1 + M)$ is exactly this **pointwise product**; losses multiply, and for shallow dips they add, $1 - |M| \simeq \sum_j (1 - |M_j|)$.



Taminiau et al., *PRL* **109**, 137602 (2012), Fig. 2(a): P_x against the interpulse delay τ , $N = 32$ pulses, $B_0 = 401$ G.

A product of responses is a convolution of coefficients

Our response is a product, $M = \prod_j M_j$, and in Fourier space a product is a convolution: one more nucleus is one more convolution of coefficient lists.

Each $M_j(\theta)$ is a function of the swept variable θ and, for K equal units, a trigonometric polynomial

$$M_j(\theta) = \sum_n c_n^{(j)} e^{in\theta} \text{ of degree } K - 1.$$

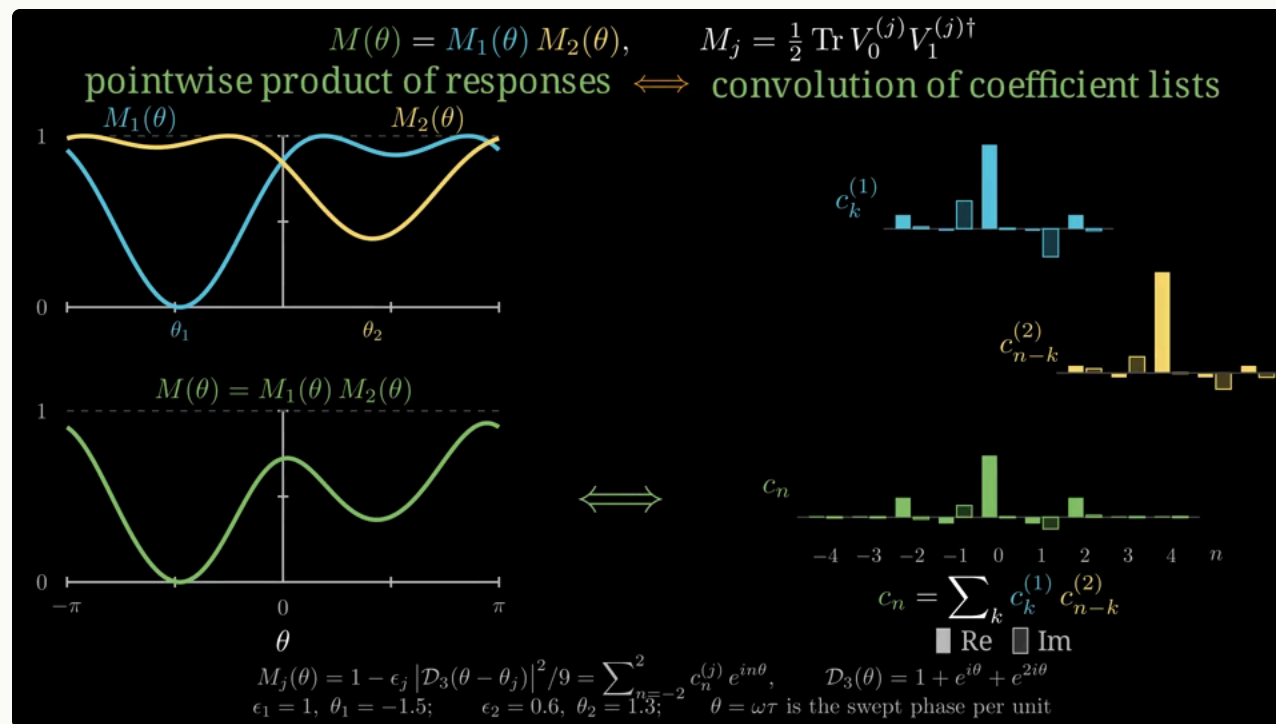
Multiplying the responses is the polynomial slide with $x = e^{i\theta}$:

$$M_1 M_2 = \sum_n c_n e^{in\theta}, \quad c_n = \sum_k c_k^{(1)} c_{n-k}^{(2)} = (c^{(1)} * c^{(2)})_n.$$

The degrees add and the lists get longer.

Chapter 3 turns this around: a QSP

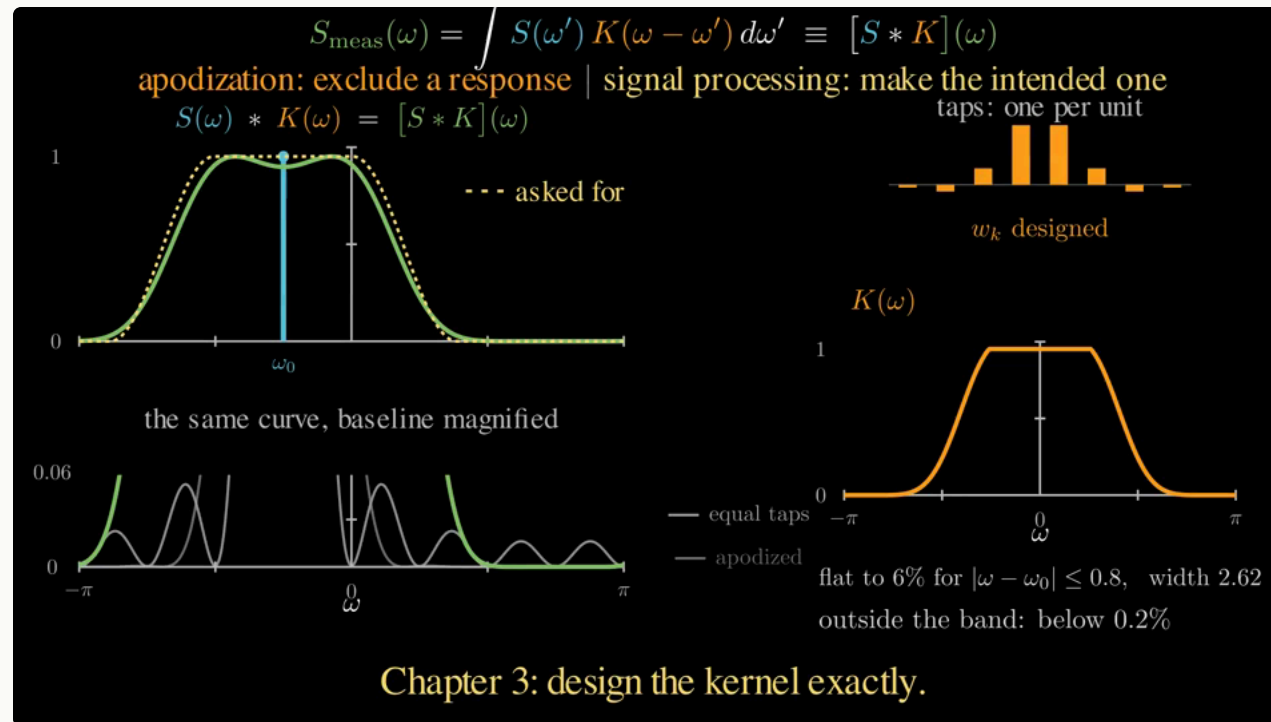
sequence multiplies degree-one factors on purpose, so it convolves its list one tap at a time.



Two nuclei give two dips $M_1(\theta)$ and $M_2(\theta)$; the measured response is their pointwise product, and its Fourier list is the convolution of the two coefficient lists (degree $2 + 2 = 4$).

What to carry out of chapter 2

- **Convolution was never a choice.** It appeared on its own in the sum of two dice, in the product of two polynomials, and in the product of many nuclear responses.
- **One factor is the response, the other is the kernel.** In $(f * g)(x) = \int f(y) g(x - y) dy$ one factor is the response of the system, the other is the kernel, also called the filter function, and which is which is our own choice.
- **So the kernel is ours to design.** A filter that excludes a part of the response we do not want is apodization, a filter that makes the response we intended is signal processing, and chapter 3 is the exact version of the second.



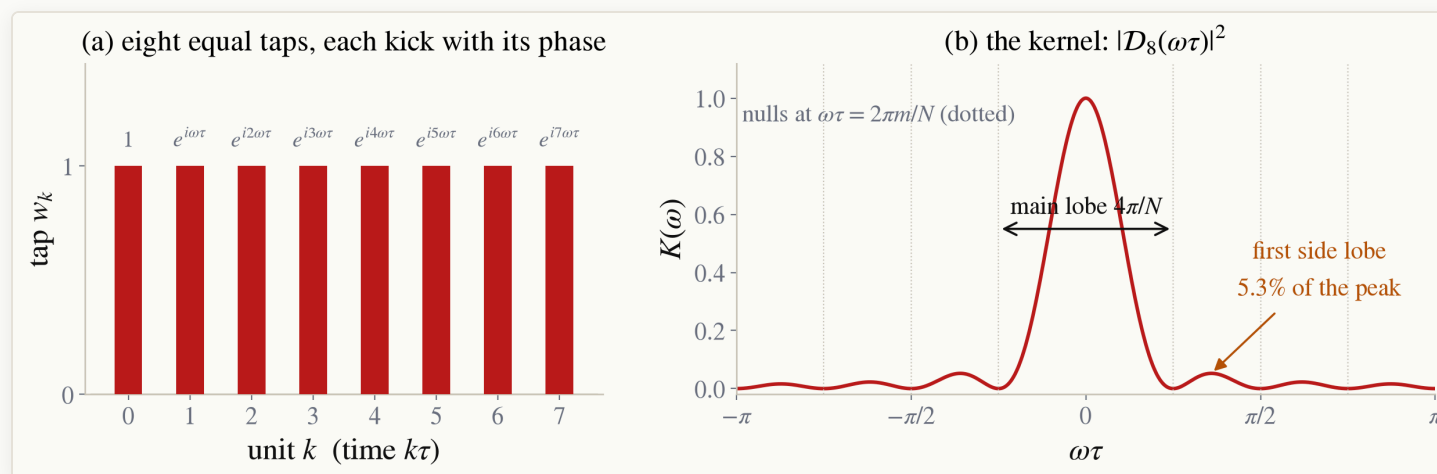
One sharp line at ω_0 read out through eight sequence units. Equal taps give the Dirichlet kernel $|\mathcal{D}_8(\omega)|^2$ with side lobes at 5.2% of the peak. A Hann taper is apodization and drops them to 0.07%, at the cost of a line 1.43 times wider. Taps designed for a band give the response we asked for.

METHOD Where the kernel comes from: the taps of the sequence

A sequence is N equal units of length τ . Unit k gives the system one small kick of amplitude w_k , the **tap**, a pulse area or an RF amplitude. Between the kicks the system precesses at the swept variable ω , so the kick of unit k arrives with the phase $e^{ik\omega\tau}$. To first order the kicks add, and the total amplitude is the Fourier series of the tap list, the polynomial of the dice slides with $z = e^{i\omega\tau}$:

$$A(\omega) = \sum_{k=0}^{N-1} w_k e^{ik\omega\tau}, \quad K(\omega) = \frac{|A(\omega)|^2}{|A(0)|^2}.$$

K is the **kernel**, or filter function: the transfer of the sequence at an offset ω from a line. Equal taps sum as a geometric series, $\sum_k e^{ik\theta} = e^{i(N-1)\theta/2} \sin(N\theta/2) / \sin(\theta/2)$, so $K = |\mathcal{D}_N(\omega\tau)|^2$ is the Dirichlet kernel.



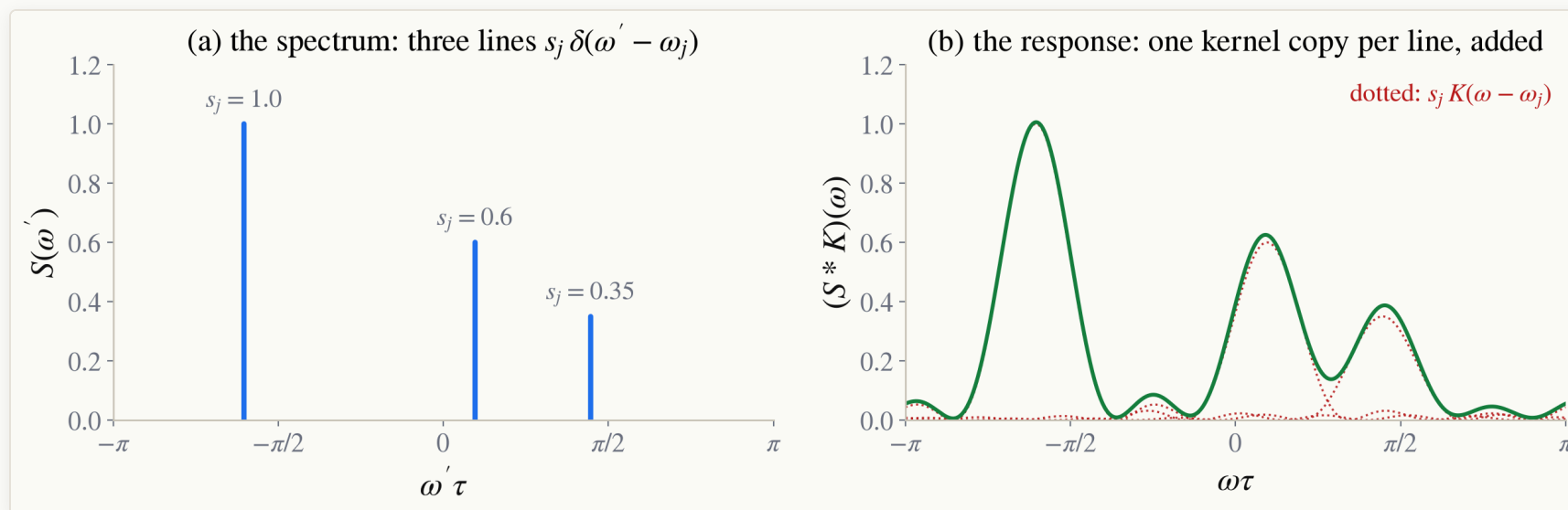
Eight equal taps and their kernel: a main lobe of width $4\pi/N$ that narrows with more units, and side lobes at 5% whatever N is.

Why the spectrum takes that form: one kernel copy per line

The sample answers at its own frequencies. Write its spectrum as a set of lines, one per source, at position ω_j with strength s_j (for a nucleus, its coupling):

$$S(\omega') = \sum_j s_j \delta(\omega' - \omega_j) \implies S_{\text{meas}}(\omega) = \int d\omega' S(\omega') K(\omega - \omega') = \sum_j s_j K(\omega - \omega_j).$$

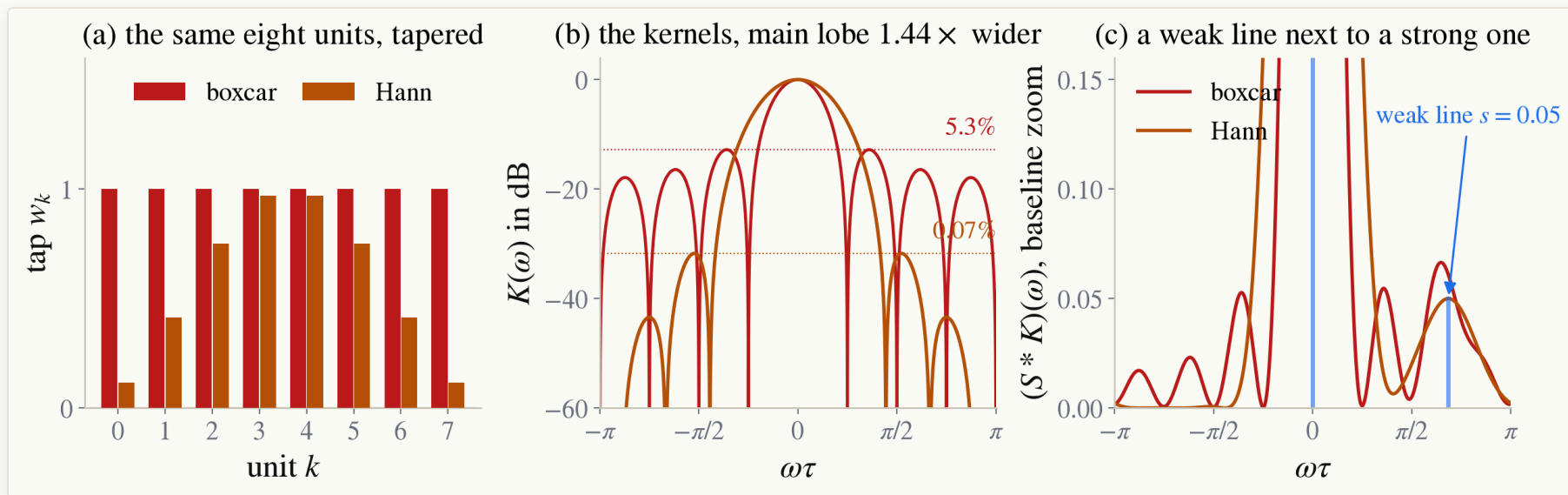
Each line sees the kernel only at its own offset $\omega - \omega_j$, so the measured curve is a copy of K placed on every line, scaled by s_j , and added. That is the convolution of the chapter opener, and it says what we can and cannot see: no line is sharper than the main lobe, and the side lobes of a strong line spill onto its neighbours.



STRATEGY

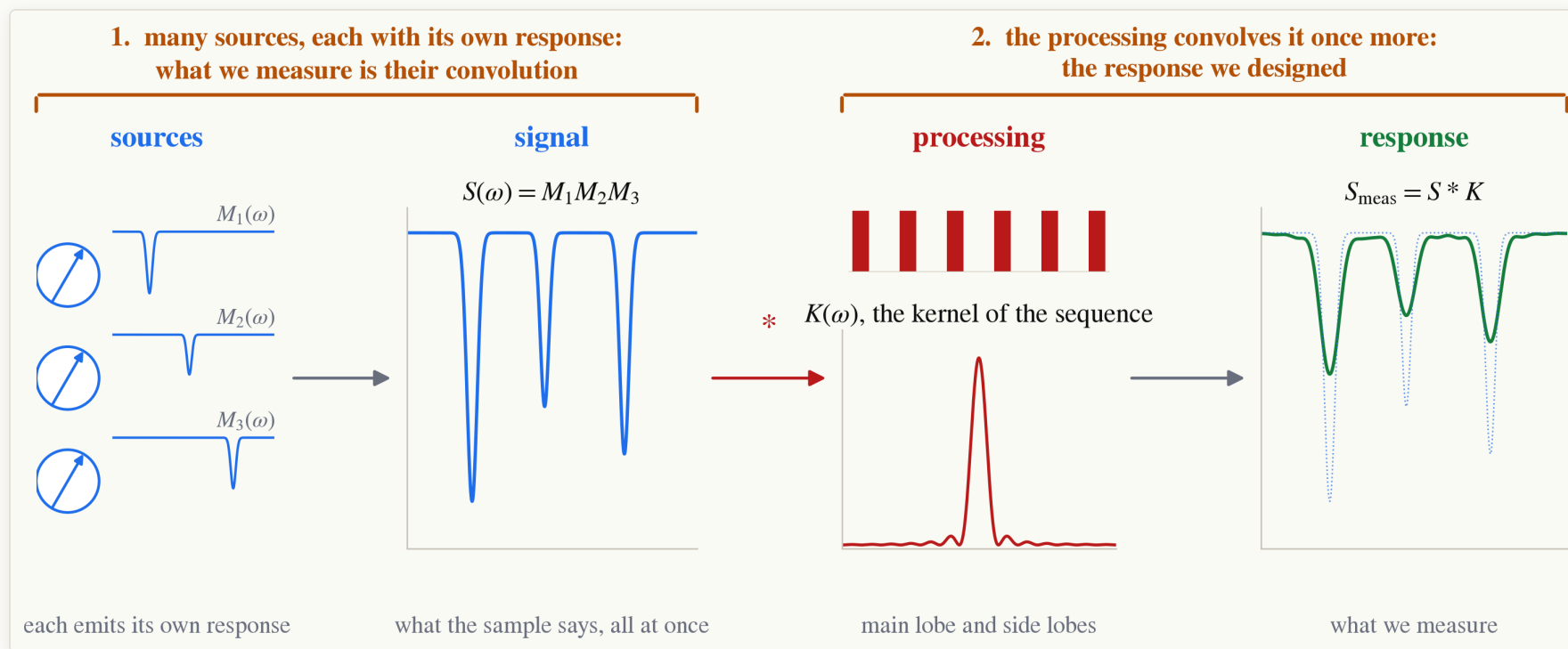
Apodization: reshape the taps, and the kernel follows

The side lobes come from the sharp edges of the boxcar: the transform of a step decays only as $1/\omega$. Taper the taps instead, $w_k = \sin^2(\pi(k+1)/(N+1))$, a Hann window. The kernel is the transform of the taps, so a smooth window gives lobes that fall away fast: for $N = 8$ the largest drops from 5.3% to 0.07% of the peak. The price comes from the same transform: a tapered list is effectively shorter, so the main lobe is 1.4 times wider. Apodization trades resolution for a clean baseline; it never makes a line narrower, only more units can.



Boxcar and Hann taps on the same eight units, their kernels in dB, and a strong line with a weak neighbour at 5% of its strength. With the boxcar the side lobes of the strong line are as tall as the weak line; with Hann only the real line remains. This is the first way to design the kernel, excluding what we do not want; chapter 3 keeps the picture, the taps are the pulses, and asks for the response we do want, exactly.

Two things are convolved



1. The sources. Many spins each emit their own response $M_j(\omega)$; the signal is their product $S = \prod_j M_j$, which is a convolution of their coefficient lists. What we measure is already a convolved signal.

2. The processing. The sequence we run is a kernel $K(\omega)$, and the read-out is the convolution $S * K$ once more. This one we design: its taps are our pulses, so it can exclude what we do not want or make the response we intended.

03

Quantum signal processing

The sequence is a polynomial, the polynomial is built by convolution, and the response is the transform of the taps

METHOD

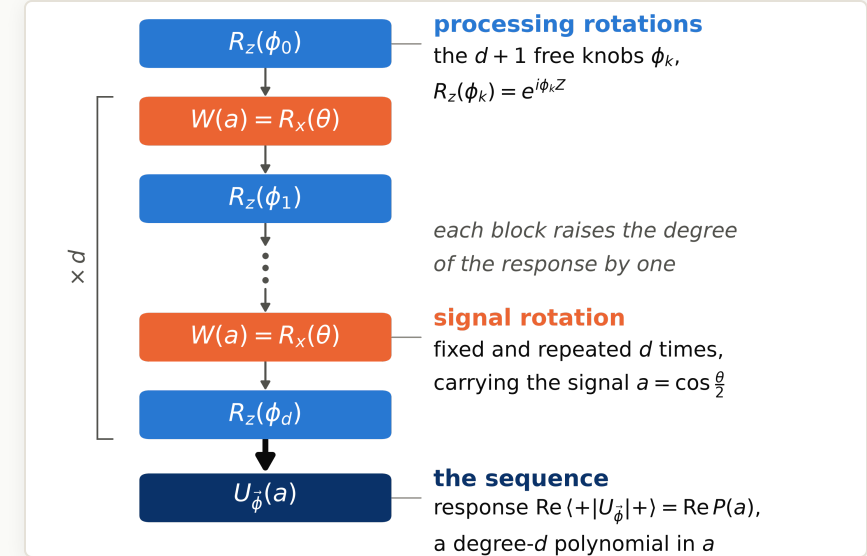
The sequence: the wait is the signal, the pulse is the knob

While the drive is off, the only surviving term is the detuning, so a wait of length τ is a precession about the z axis,

$$W(\delta) = R_z(\delta\tau) = \begin{pmatrix} z^{-1/2} & 0 \\ 0 & z^{1/2} \end{pmatrix}, \quad z = e^{i\delta\tau}.$$

It is the **signal**: the detuning δ is what the system hands us and W is the only place it enters. The microwave pulses G_k are the **processing**, chosen freely through their area β_k and carrier phase φ_k . With d equal waits and $d + 1$ pulses,

$$U(\delta) = G_d W(\delta) G_{d-1} \cdots W(\delta) G_0.$$



The textbook ladder of [1] puts the signal in an x -rotation $W_x(a)$ and the processing in z -rotations. It is the same sequence in a rotated basis, since $H W_x(a) H$ is our wait with $a = \cos(\delta\tau/2)$ and $H e^{i\phi_k Z} H = R_x(-2\phi_k)$ is our pulse.

METHOD Every factor is degree one, so the sequence is a polynomial

Split the half-powers off the wait,
 $W(\delta) = z^{-1/2} A(z)$ with $A(z) = \text{diag}(1, z)$,
 so the d of them make one phase:

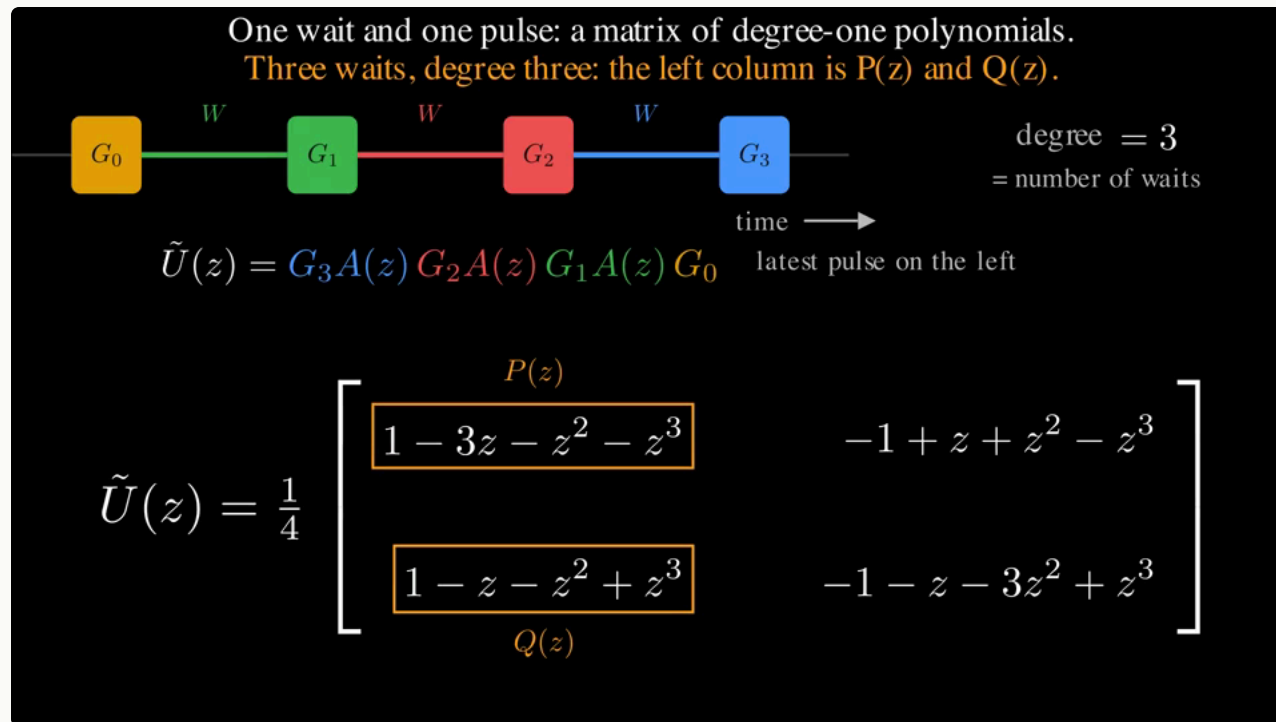
$$U(\delta) = z^{-d/2} \tilde{U}(z),$$

$$\tilde{U}(z) = G_d A(z) G_{d-1} A(z) \cdots G_1 A(z) G_0.$$

Each coloured block is one wait then one pulse: a 2×2 matrix of degree-one polynomials in z . **Degree = number of waits.** Every G_k and every W is in $SU(2)$, so

$$U(\delta) = \begin{pmatrix} \hat{P} & -\hat{Q}^* \\ \hat{Q} & \hat{P}^* \end{pmatrix}, \quad |\hat{P}|^2 + |\hat{Q}|^2 = 1.$$

Here $\hat{P} = z^{-d/2} P(z)$ and $\hat{Q} = z^{-d/2} Q(z)$ with $\deg P, \deg Q \leq d$. We prepare $|0\rangle$ and read $|0\rangle$, so the experiment measures exactly $|\hat{P}(\delta)|^2$: **designing the sequence means designing one polynomial.**



Four pulses $R_y(\pi/2)$ and three waits, one coloured block at a time: every wait raises the degree by one, and the left column of $\tilde{U}(z)$ is $P(z)$ over $Q(z)$.

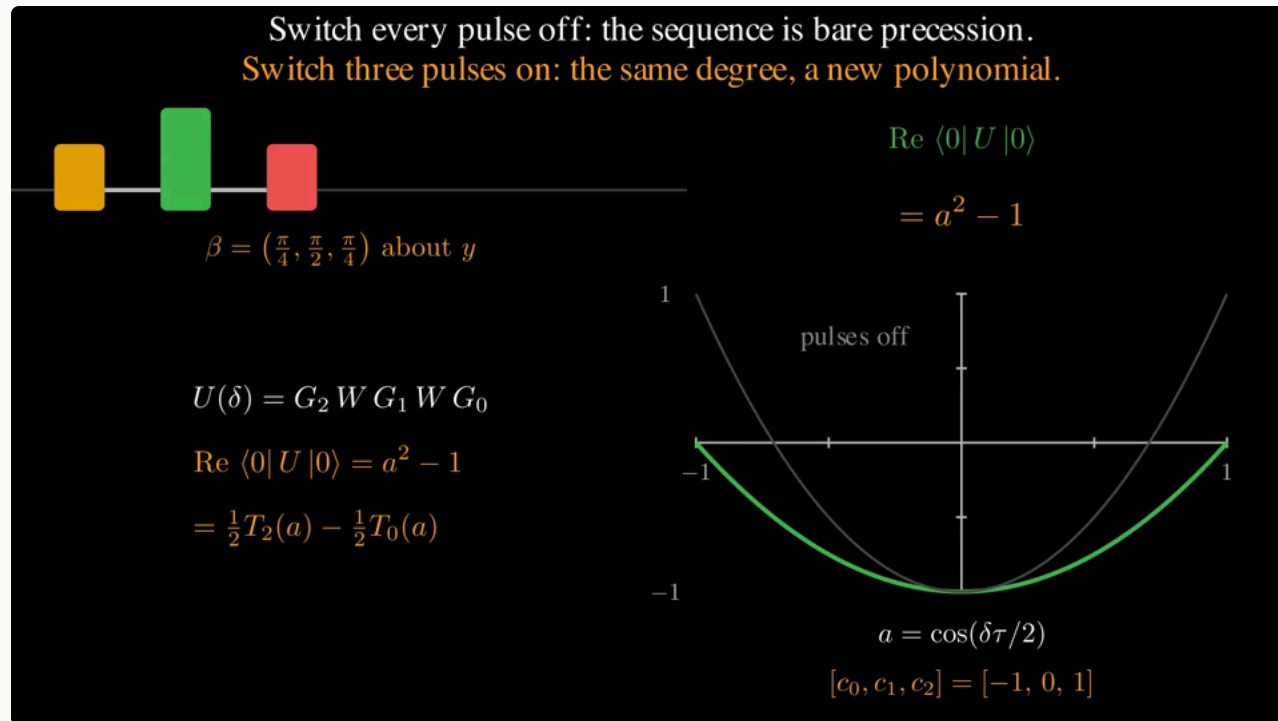
A concrete example: what a polynomial response means

Switch every pulse off. The sequence is bare precession, $U(\delta) = W(\delta)^d$, and the amplitude we read is one pure harmonic:

$$\langle 0|U|0\rangle = e^{-i d \delta\tau/2},$$

$$\text{Re } \langle 0|U|0\rangle = \cos\left(d \frac{\delta\tau}{2}\right) = T_d(a),$$

with $a = \cos(\delta\tau/2)$: at $d = 1$ that is a , at $d = 2$ it is $2a^2 - 1$, at $d = 5$ it is $16a^5 - 20a^3 + 5a$. These are the Chebyshev polynomials, the textbook's $P = T_d(a)$ and $Q = U_{d-1}(a)$.



Pulses off at $d = 1, 2, 5$, then on at $d = 2$: same degree, new polynomial.

Now switch three pulses on at $d = 2$, areas $\beta = (\frac{\pi}{4}, \frac{\pi}{2}, \frac{\pi}{4})$ about y : the same read-out becomes $\text{Re } \langle 0|U|0\rangle = a^2 - 1 = \frac{1}{2}T_2(a) - \frac{1}{2}$, a different polynomial of the same degree. **The waits fix the degree, the pulses choose which polynomial of that degree we measure.**

One wait and one pulse are a two-tap kernel

Let $\Pi_b = |b\rangle\langle b|$ be the projector onto $|b\rangle$, so that $\Pi_0 + \Pi_1 = I$. During a wait only $|1\rangle$ collects the phase $\delta\tau$ relative to $|0\rangle$, so the wait leaves the $|0\rangle$ component where it is and advances the $|1\rangle$ component by one power of z :

$$A(z) = \text{diag}(1, z) = \Pi_0 + z\Pi_1, \quad z = e^{i\delta\tau}.$$

One power of z is one wait. The pulse that follows is a fixed matrix while z is a scalar, so it multiplies straight through both terms:

$$G_k A(z) = K_k^{(0)} + z K_k^{(1)}, \quad K_k^{(0)} = G_k \Pi_0, \quad K_k^{(1)} = G_k \Pi_1.$$

Multiplying by Π_0 keeps the first column of G_k and deletes the second, and Π_1 does the reverse. For a pulse about x of area β_k , writing $c = \cos \frac{\beta_k}{2}$ and $s = \sin \frac{\beta_k}{2}$,

$$G_k = R_x(\beta_k) = \begin{pmatrix} c & -is \\ -is & c \end{pmatrix}, \quad K_k^{(0)} = \begin{pmatrix} c & 0 \\ -is & 0 \end{pmatrix}, \quad K_k^{(1)} = \begin{pmatrix} 0 & -is \\ 0 & c \end{pmatrix}.$$

Each step of the sequence is therefore a **two-tap kernel whose taps are matrices**. The tap at z^0 sends the $|0\rangle$ amplitude through the pulse at the same power of z , and the tap at z^1 sends the $|1\rangle$ amplitude through the pulse one power of z later.

METHOD

Each added step convolves the coefficient list

The state after k steps is a list of 2-vectors, one entry per number of waits:

$$\tilde{U}_k|0\rangle = \sum_n v_n^{(k)} z^n.$$

The coefficient of z^n in $G_k A(z) \tilde{U}_{k-1}$ is

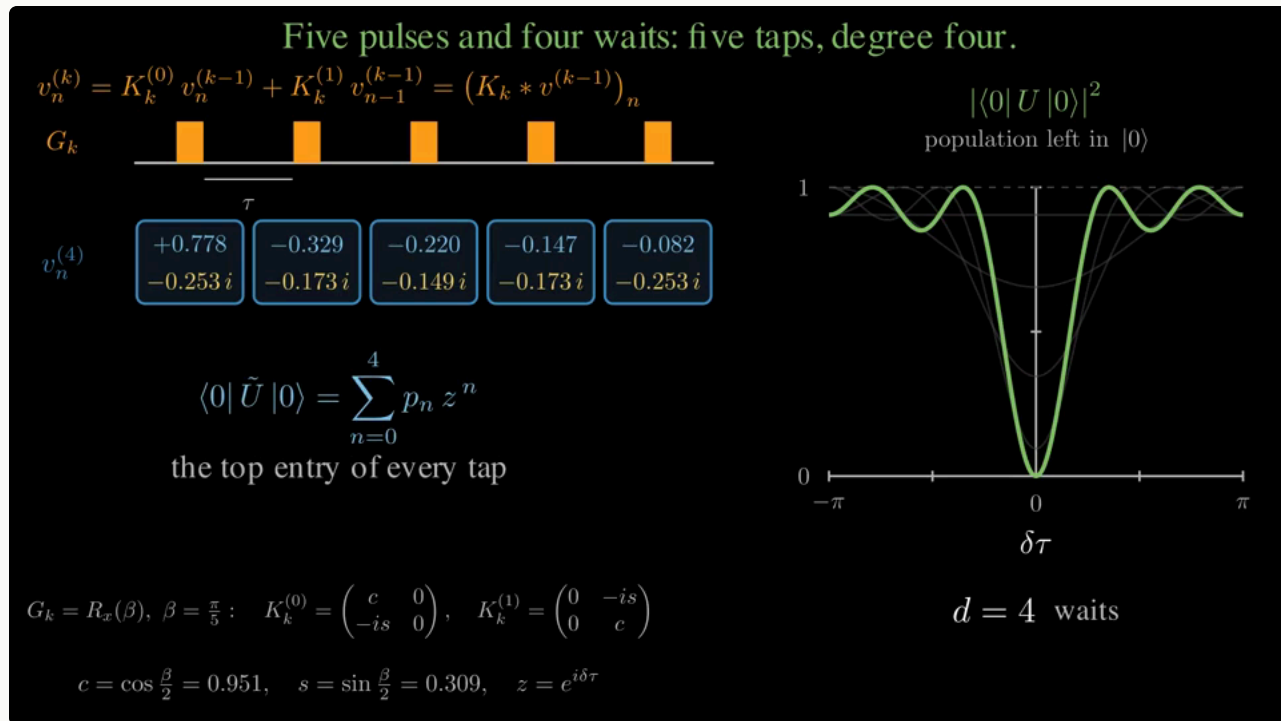
$$v_n^{(k)} = K_k^{(0)} v_n^{(k-1)} + K_k^{(1)} v_{n-1}^{(k-1)},$$

a flip-and-slide sum with matrix taps, $(K_k * v^{(k-1)})_n$.

Unrolled from $v_0^{(0)} = G_0|0\rangle$:

$$\begin{aligned} v_0^{(1)} &= K_1^{(0)} v_0^{(0)}, & v_1^{(1)} &= K_1^{(1)} v_0^{(0)}, \\ v_0^{(2)} &= K_2^{(0)} v_0^{(1)}, & v_2^{(2)} &= K_2^{(1)} v_1^{(1)}, \\ v_1^{(2)} &= K_2^{(0)} v_1^{(1)} & &+ K_2^{(1)} v_0^{(1)}. \end{aligned}$$

At either end the kernel hangs off the list and only one term survives, so each step adds exactly one entry:
the degree is the number of waits.



Five pulses $G_k = R_x(\pi/5)$, whose areas add to π , and four waits. The reversed kernel $(K_k^{(1)}, K_k^{(0)})$ slides along the old list from $n = 0$ to $n = k$, and each stop writes one entry of the new list; the printed numbers are the entries of $v_n^{(k)}$. On the right the response $|\langle 0 | U | 0 \rangle|^2$ over $\delta\tau$ gains one extremum per degree.

The response is the Fourier transform of the taps

Keep only the $|0\rangle$ component of the list. Writing p_n for the upper entry of v_n ,

$$P(z) = \langle 0 | \tilde{U} | 0 \rangle = \sum_{n=0}^d p_n z^n.$$

The experiment does not hand us a free z : the sweep puts z on the unit circle, $z = e^{i\delta\tau}$. Restoring the phase $z^{-d/2}$ split off the waits,

$$\hat{P}(\delta) = z^{-d/2} P(z) = \sum_{n=0}^d p_n e^{i(n-\frac{d}{2})\delta\tau},$$

a **Fourier series in the detuning** whose coefficients are the taps. Each tap is one harmonic of the wait: an amplitude that has waited n times carries the phase $n\delta\tau$, so p_n oscillates in δ at the frequency $n\tau$.

The measurement squares it, and $|\hat{P}|^2 = \hat{P}\hat{P}^*$ is a product of two such series. By the one identity of Section 2 a product of series is a convolution of lists, so

$$|\hat{P}(\delta)|^2 = \sum_{m=-d}^d r_m e^{im\delta\tau}, \quad r_m = \sum_n p_{n+m} p_n^*,$$

and the measured curve is the transform of the **autocorrelation** of the tap list.

Take $d = 1$, two taps. Then

$$|\hat{P}|^2 = |p_0|^2 + |p_1|^2 + 2 \operatorname{Re}(p_1 p_0^* e^{i\delta\tau}),$$

a constant plus one cosine. Two $\pi/2$ pulses of opposite sign around one wait give $p_0 = p_1 = \frac{1}{2}$ and $\frac{1}{2}(1 + \cos \delta\tau)$, the Ramsey fringe of Section 1.

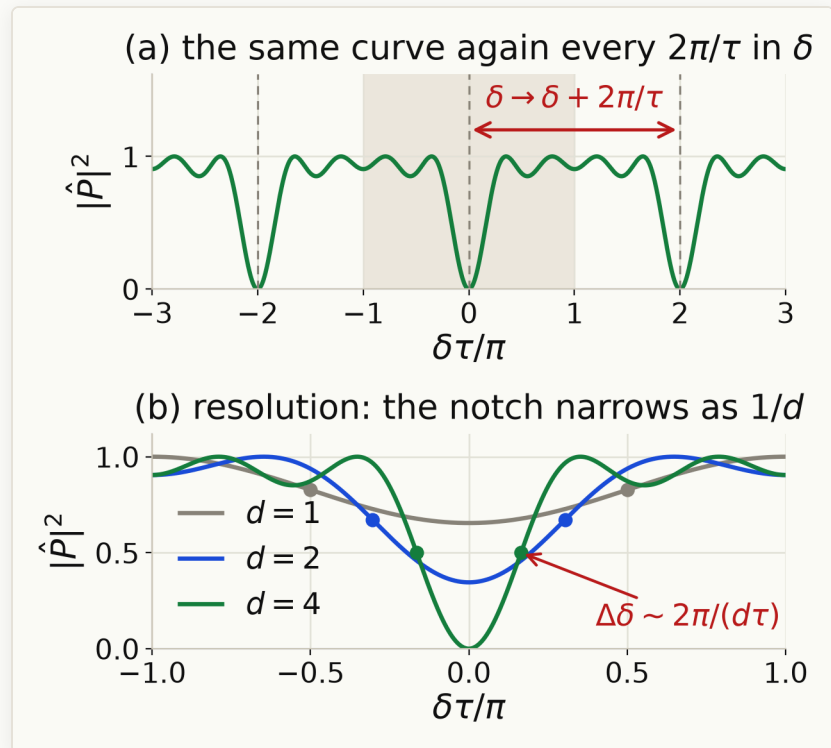
Two numbers follow: the period and the resolution

Periodicity. The response depends on the detuning only through $z = e^{i\delta\tau}$, and z returns to itself when $\delta\tau$ advances by 2π , so the measured curve repeats with period $2\pi/\tau$ in δ . The wait is a **sampling interval**: choosing τ chooses where the lines of the spectrum land on the circle, and two lines a multiple of $2\pi/\tau$ apart are aliased onto the same point and can never be told apart.

Resolution. The highest harmonic on the list is $e^{\pm id\delta\tau/2}$, so the fastest the curve can vary is set by $d\tau$ and the narrowest feature it can carry is

$$\Delta\delta \sim \frac{2\pi}{d\tau} = \frac{2\pi}{T}.$$

The total sequence time $T = d\tau$ is a **record length**, and this is the statement Section 2 made with the Dirichlet kernel: a record of finite length convolves every line with the transform of its window. **Resolution is bought with sequence length.**



The same sequence as the clip, $G_k = R_x(\pi/5)$ with d equal waits. Top: the $d = 4$ curve over three periods, the shaded band being one period. Bottom: the notch at $d = 1, 2, 4$, with dots at half its depth. Fig. by `fig_response.py`, 2026-09-22.

Weak pulses: the taps are the pulse amplitudes

Expand each pulse to first order,

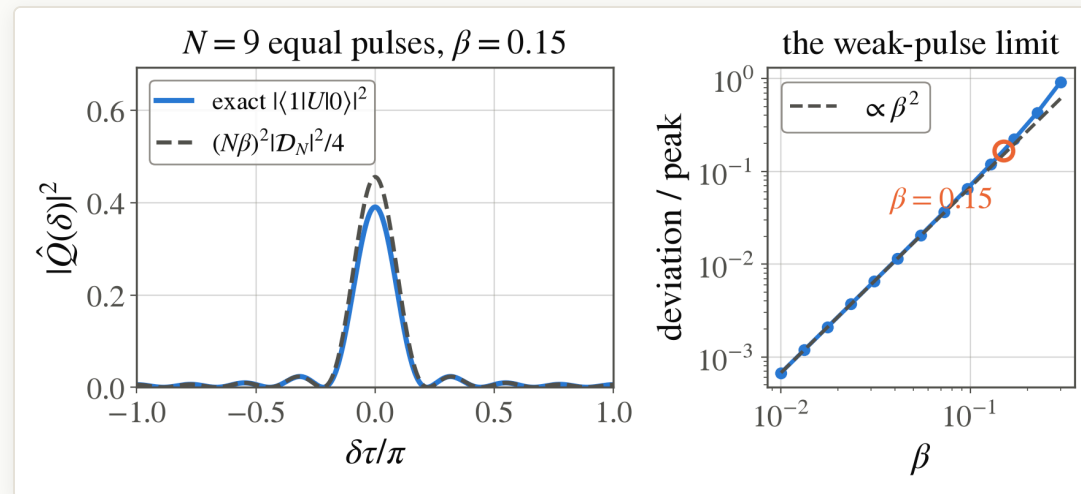
$G_k \simeq I - \frac{i}{2} \beta_k (\cos \varphi_k X + \sin \varphi_k Y)$, and keep the terms with one pulse in them. Each wait left of pulse k contributes $e^{+i\delta\tau/2}$ and each wait to its right $e^{-i\delta\tau/2}$, so

$$\hat{Q}(\delta) \simeq -\frac{i}{2} \sum_{k=0}^d \beta_k e^{i\varphi_k} e^{i(\frac{d}{2}-k)\delta\tau}.$$

The tap q_k is the complex amplitude $\beta_k e^{i\varphi_k}$: the knobs on the bench are the coefficient list. Equal pulses are the all-ones boxcar of length $N = d + 1$, and its transform is the Dirichlet kernel,

$$|\hat{Q}(\delta)|^2 = \frac{(N\beta)^2}{4} |\mathcal{D}_N(\delta\tau)|^2, \quad \mathcal{D}_N(\theta) = \frac{\sin(N\theta/2)}{N \sin(\theta/2)},$$

the instrument function of Section 2.

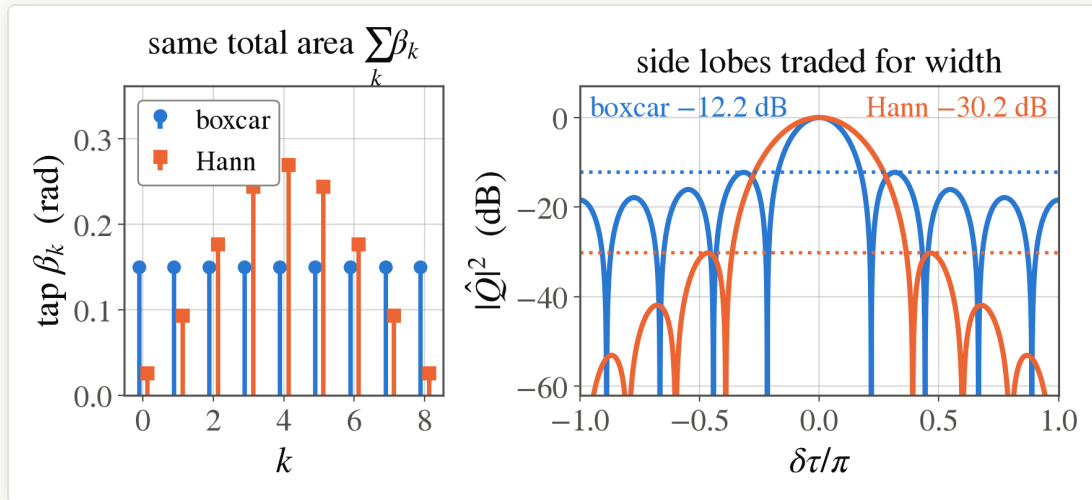


Left: the transfer of $N = 9$ equal pulses of area $\beta = 0.15$, from the exact product $U = G_d W \cdots W G_0$ (solid) and from the formula above (dashed).

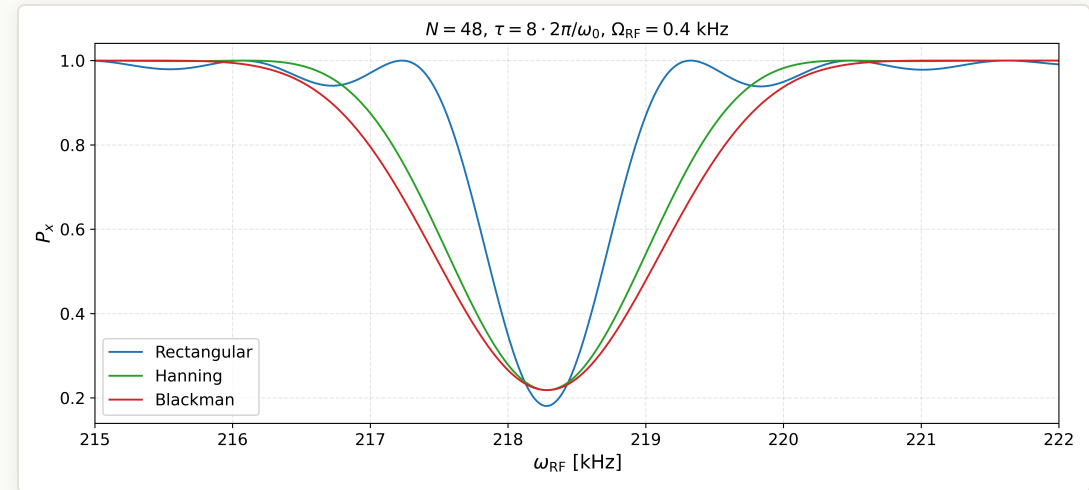
Every null and every side lobe coincides, and only the peak differs, $\sin^2(N\beta/2) = 0.391$ against $(N\beta)^2/4 = 0.456$. Right: that largest deviation is 17% of the peak here, falls as β^2 , and is 1.7% at $\beta = 0.05$. From our own evaluator, `fig_qsp_design.py`.

RESULTS

Shaping the areas is apodization, in our data too



Nine pulses with the same total area $\sum_k \beta_k = 1.35$, hence the same transfer on resonance. A Hann envelope pushes the largest side lobe from -12.2 dB down to -30.2 dB, a factor of 63 in transferred population, and pays for it with a main lobe 1.46 times wider. Both curves are exact products, which is why the boxcar lobe sits 0.7 dB above the -12.9 dB of the first-order window.



The same trade in our own DDrf data. The taps are the amplitudes $\Omega_k = \Omega f(k)$ of the $N = 48$ RF cells, the swept variable is the RF frequency ω_{RF} , and P_x is the surviving population, so $1 - P_x$ is the transfer that plays the role of $|\hat{Q}|^2$. A rectangular envelope leaves ripples of a few percent beside the line; Hanning and Blackman remove them and widen the dip. (source: lab meeting 2026-04-28)

Shaping the β_k is apodization, so in this limit QSP is classical window design. At large pulse areas the linear map from amplitudes to taps is replaced by the exact matrix convolution we started from, and that is where QSP goes further.

The QSP theorem: the reachable responses are known exactly

Theorem. With d waits and $d + 1$ pulses the sequence is always $U = \begin{pmatrix} \hat{P} & -\hat{Q}^* \\ \hat{Q} & \hat{P}^* \end{pmatrix}$ with $\deg P, \deg Q \leq d$ and $|\hat{P}|^2 + |\hat{Q}|^2 = 1$ on the circle, and conversely **every** such pair is realized by some choice of pulses [2, Thm. 3]. The textbook W_x form is the restricted version, with real coefficients and definite parity: $\deg P \leq d, \deg Q \leq d - 1$, parities d and $d - 1$, and $|P|^2 + (1 - a^2)|Q|^2 = 1$ [1, Thm. 1].

The forward direction is nothing but the convolution recursion: each step appends one tap and raises the degree by one, which is what flips the parity in the W_x form, while unitarity is inherited because every factor is unitary on the circle. The converse is the direction we use, because it says the design space is a **complete** list rather than a perturbative family: any bounded polynomial of degree d is a sequence we can actually build, so a polynomial that vanishes exactly at three chosen detunings is a gate that is exactly blind to three spectral lines.

What P is with no pulses, and what it can be with them

Switch every pulse off. The sequence is bare free precession, $U = W(\delta)^d$, whose tap list has a single nonzero entry, so its response is one pure harmonic,

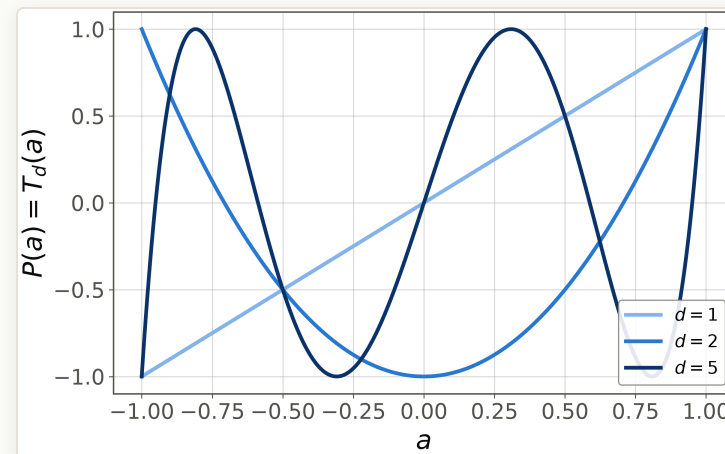
$$\langle 0|U|0\rangle = e^{-id\delta\tau/2}, \quad \text{Re } \langle 0|U|0\rangle = \cos(dx) = T_d(a),$$

with $a = \cos x$ and $x = \delta\tau/2$. In the textbook labelling this is the trivial-phase case of [1], $P = T_d(a)$ and

$Q = U_{d-1}(a)$, Chebyshev of the first and second kind, and the unitarity condition is just

$$T_d^2 + (1 - a^2)U_{d-1}^2 = \cos^2(dx) + \sin^2(dx) = 1.$$

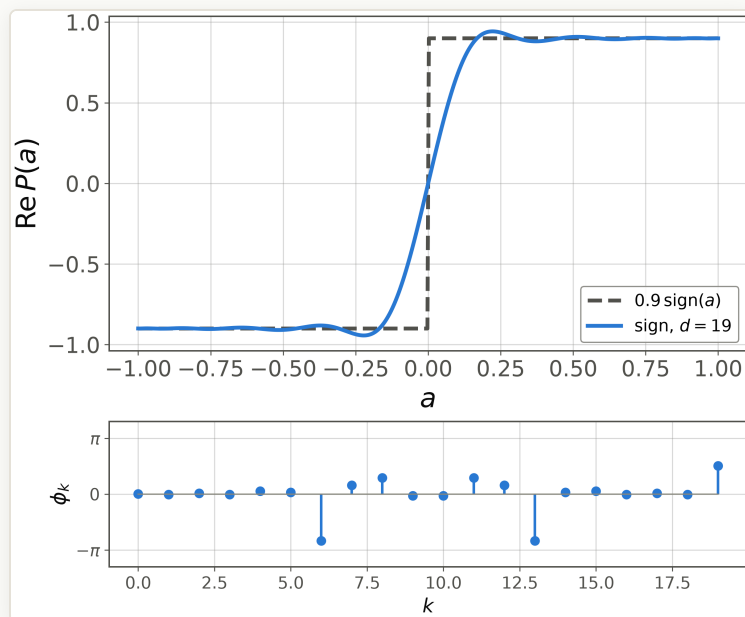
Chebyshev polynomials are the pure harmonics of the wait, and the pulses are what turns one harmonic into a designed sum of them.



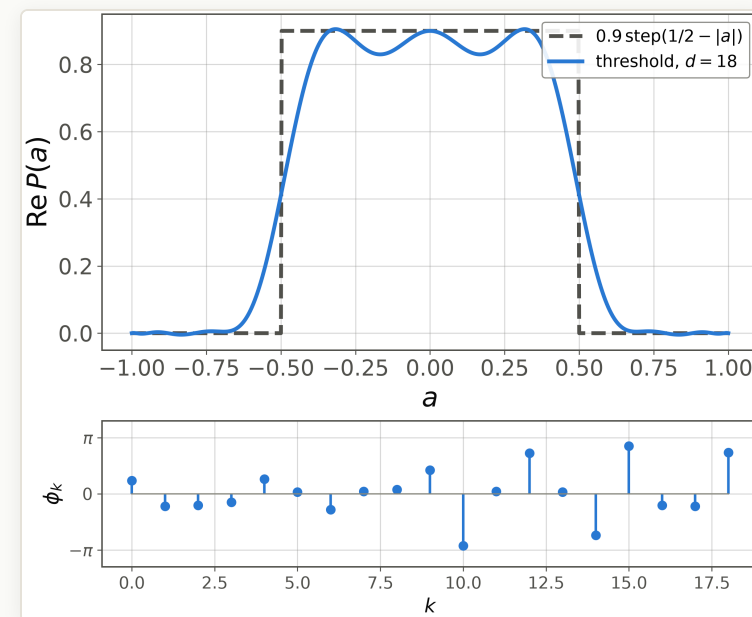
$P(a) = T_d(a)$ at trivial phases for $d = 1, 2, 5$, from our evaluator (agrees to 10^{-12}).

At $a = \pm 1$ the wait is trivial, so the sequence collapses to one fixed rotation and $|P(\pm 1)| = 1$ is forced: a target such as $\frac{1}{2} \text{sign}(a)$ is unreachable as P itself. Reading only the real part cures it, since $\text{Im } P$ can absorb the unit-modulus burden. **Re P may then be any real polynomial with parity $d \bmod 2$ and $|\text{Re } P| \leq 1$ [1].**

Choosing the polynomial is choosing the algorithm



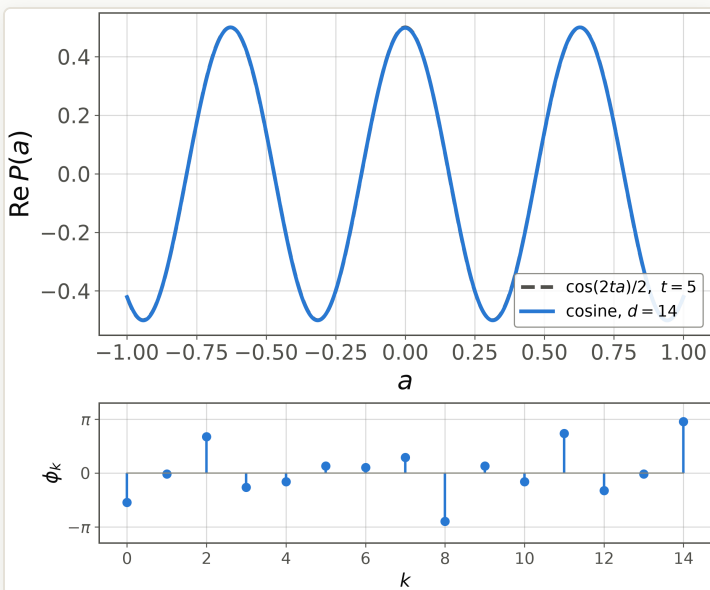
Sign function at $d = 19$, the decision primitive behind amplitude amplification and search.



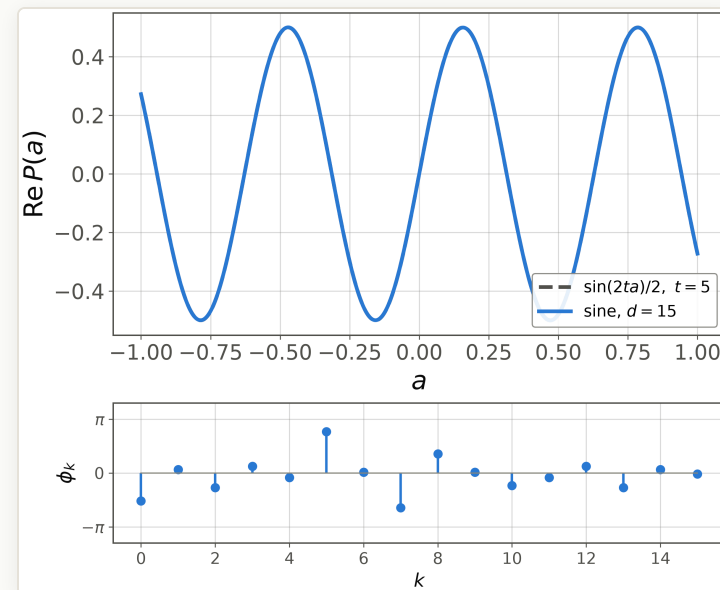
Threshold function at $d = 18$, a band-pass response used for eigenvalue thresholding and eigenstate filtering.

The lower panel of each figure is the phase list, which is the entire program. Appendix D of [1] gives lists for $\cos$, $\sin$, $1/a$ and $e^{-\beta a}$ as well.

Cosine and sine: one tap list is a simulation program



Cosine response at $t = 5$, $d = 14$, from the phase list of [1, App. D4].



Sine response at $t = 5$, $d = 15$, from the phase list of [1, App. D4].

Hamiltonian simulation uses the pair through $e^{-i\mathcal{H}t} = \cos(\mathcal{H}t) - i \sin(\mathcal{H}t)$, so these two angle lists are the entire program, and the degree grows linearly with the simulated time t .

From a target to the pulses: first the partner

Everything so far designed a polynomial. The bench needs the $2(d+1)$ numbers β_k and φ_k , so the design is finished only once P is turned back into pulses.

Complete the matrix first: unitarity on the circle asks for a partner with $|Q|^2 = 1 - |P|^2$, and the bound $|P| \leq 1$ is the only obstruction, because no entry of a unitary matrix can exceed one in modulus.

Fejér and Riesz: a trigonometric polynomial that is non-negative everywhere on the circle is the squared modulus of an ordinary polynomial of the same degree.

Since $S = 1 - |P|^2 \geq 0$ is exactly such a polynomial, the partner exists. To build it, note that $z^d S(z)$ has $2d$ roots which pair up as $(r, 1/\bar{r})$ because S is real on the circle, and keeping one root of each pair fixes Q [2].

With P and Q in hand every tap of $\tilde{U}(z) = \sum_n C_n z^n$ is known. Its two ends are single terms, since only one choice of projector per wait reaches an extreme power:

$$\begin{aligned} C_d &= G_d \Pi_1 G_{d-1} \Pi_1 \cdots \Pi_1 G_0, \\ C_0 &= G_d \Pi_0 G_{d-1} \Pi_0 \cdots \Pi_0 G_0. \end{aligned}$$

Each $\Pi_b = |b\rangle\langle b|$ cuts its product into scalars,

$$C_d = \left(\prod_{k=1}^{d-1} \langle 1|G_k|1\rangle \right) G_d|1\rangle\langle 1|G_0,$$

so both ends have rank one, with column spaces $G_d|1\rangle$ and $G_d|0\rangle$. Unitarity makes those orthogonal: on the circle $\tilde{U}^\dagger = \sum_n C_n^\dagger z^{-n}$, so $\tilde{U}\tilde{U}^\dagger = I$ reads coefficient by coefficient as $\sum_n C_{n+m} C_n^\dagger = \delta_{m0} I$, whose top power $m = d$ is the single term $C_d C_0^\dagger = 0$. **The last pulse is read off the two ends of the list.**

Peeling the pulses off is deconvolution

Now strip G_d away. In

$G_d^\dagger \tilde{U}(z) = A(z)G_{d-1} \cdots A(z)G_0$ the top tap carries Π_1 on the left and the bottom tap carries Π_0 , so the $\langle 0|$ row has no z^d and the $\langle 1|$ row no z^0 . Dividing that row by z is legal, and

$$A(z)^{-1} G_d^\dagger \tilde{U}(z) = G_{d-1} A(z) \cdots A(z) G_0$$

is the same object with one wait fewer, so d steps leave G_0 . This is literally deconvolution: one two-tap kernel $G_k \Pi_0 + z G_k \Pi_1$ per step built the sequence, and the recursion divides one out per step, in reverse order.

One wait, two pulses. With $G_0 = R_x(\pi/3)$ and $G_1 = R_x(\pi/2)$, $\tilde{U}(z) = C_0 + zC_1$ where

$$C_0 = \frac{1}{4} \begin{pmatrix} \sqrt{6} & -i\sqrt{2} \\ -i\sqrt{6} & -\sqrt{2} \end{pmatrix}, \quad C_1 = \frac{1}{4} \begin{pmatrix} -\sqrt{2} & -i\sqrt{6} \\ -i\sqrt{2} & \sqrt{6} \end{pmatrix}.$$

Both determinants vanish and $C_1 C_0^\dagger = 0$ exactly. The column directions $(1, -i)/\sqrt{2}$ and $(1, i)/\sqrt{2}$ are orthogonal as promised, and as the two columns of one $SU(2)$ matrix they rebuild $G_1 = R_x(\pi/2)$, up to one phase per column that is a harmless virtual z -rotation. One division then returns G_0 , and the same recursion peels a random $d = 3$ sequence back to 10^{-15} .

Optimization over the areas and phases skips the factorization altogether, the route of `pyqsp` and QSPACK and of our own fits of 2026-08-18, at the price of local minima. For a **gate** target the design is instead a linear program in P with a global optimum, whose pulses this same peel-off returns [3].

Summary: three sections, one sentence

A quantum experiment prepares a state, applies a unitary and measures; the unitary is what we design, the measurement returns a response, and that response is a convolution that QSP lets us design exactly.

- Section 1 fixed the vocabulary: a state, a unitary built from a drift term and a control term, and a projective readout whose statistics are the response.
- Section 2 showed that responses multiply and therefore convolve, from the product over nuclear spins to the Dirichlet instrument function of a decoupling spectrum.
- Section 3 made the convolution the design variable. With d taps the sequence is a degree- d polynomial in $z = e^{i\delta\tau}$ assembled by d two-tap convolutions, the measured curve is the Fourier transform of that tap list, weak pulses reproduce classical windowing, and the theorem says every bounded polynomial is reachable with pulses recovered by a peel-off recursion.
- The concrete payoff is the conditional gate on the V_B^- centre, where a seven-pulse sequence is exactly the identity on three hyperfine lines and $R_x(\pi)$ on the fourth [3].

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Thank you

Questions & discussion

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